Yandex Research

Evolution of Conditioning and Multi-modal Generation

Sergey Kastryulin

Agenda

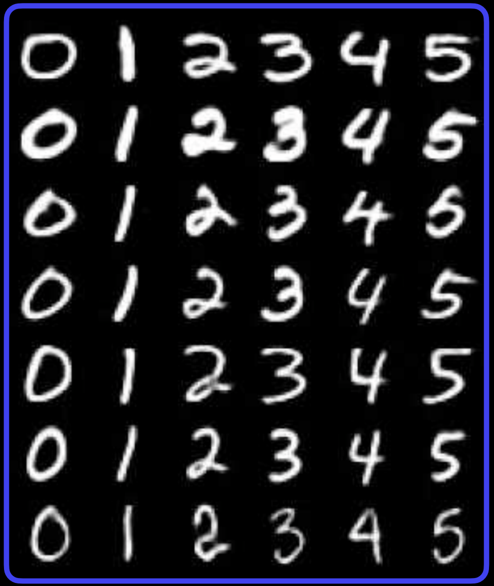
01 Basic conditioning

02 Text and Image conditioning

03 Multi-modal generation

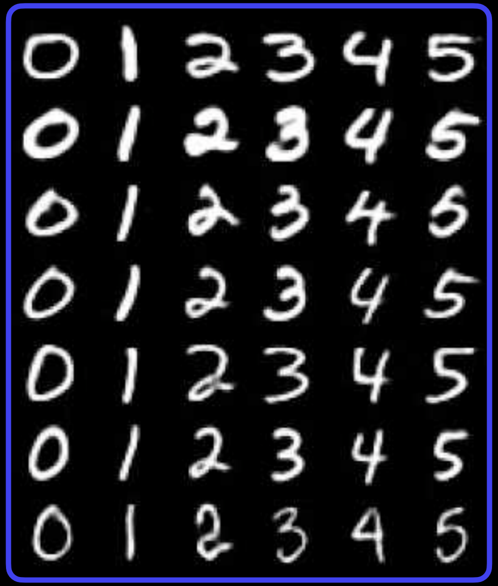
04 Inference-time compute scaling

Basic Conditioning

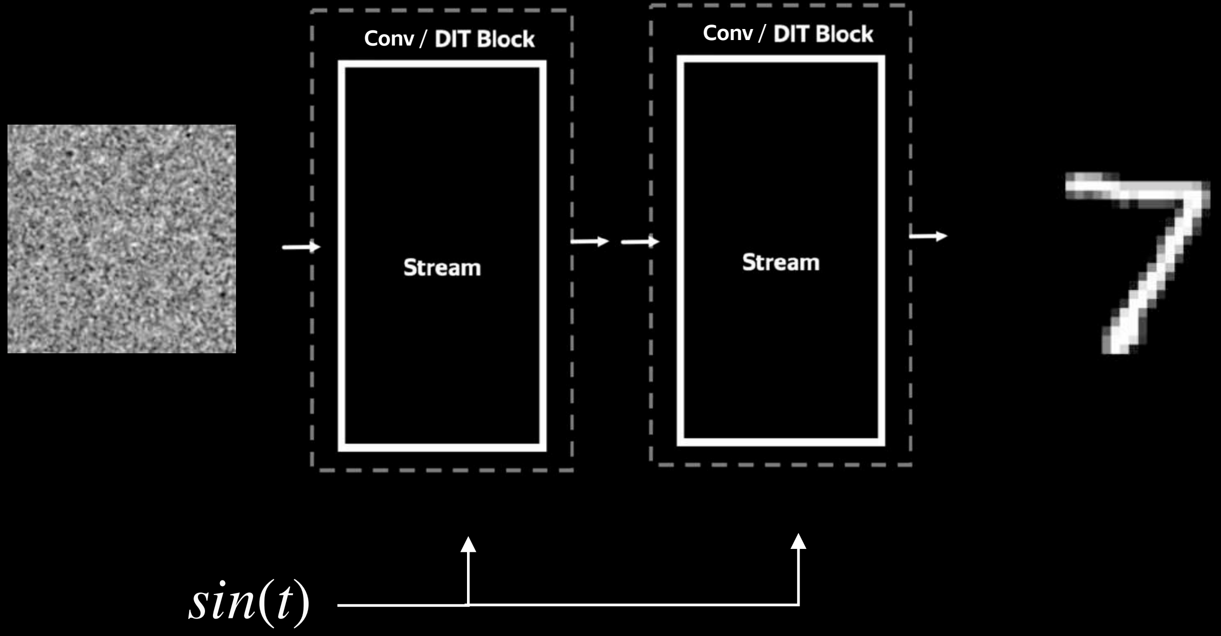


Unconditional

Basic Conditioning



Unconditional

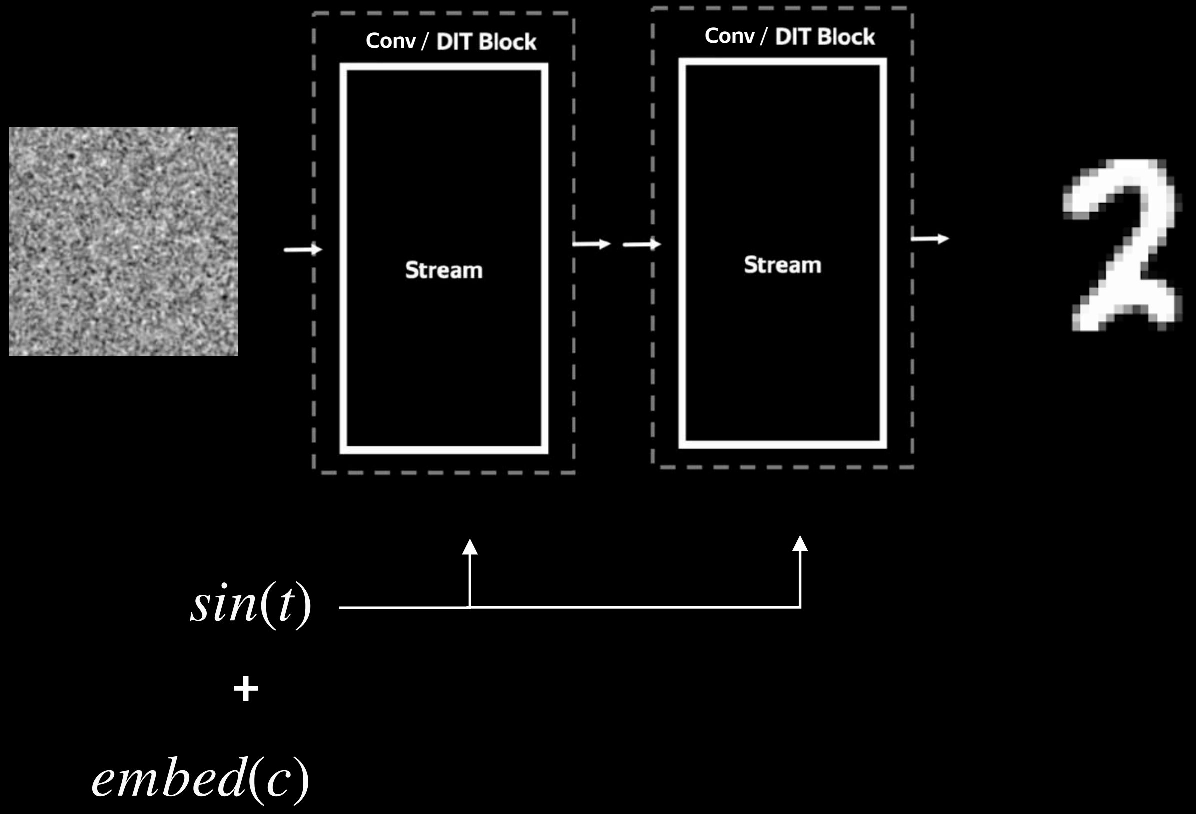


Basic Conditioning

Class-conditional

Basic Conditioning

Class-conditional



Basic Conditioning

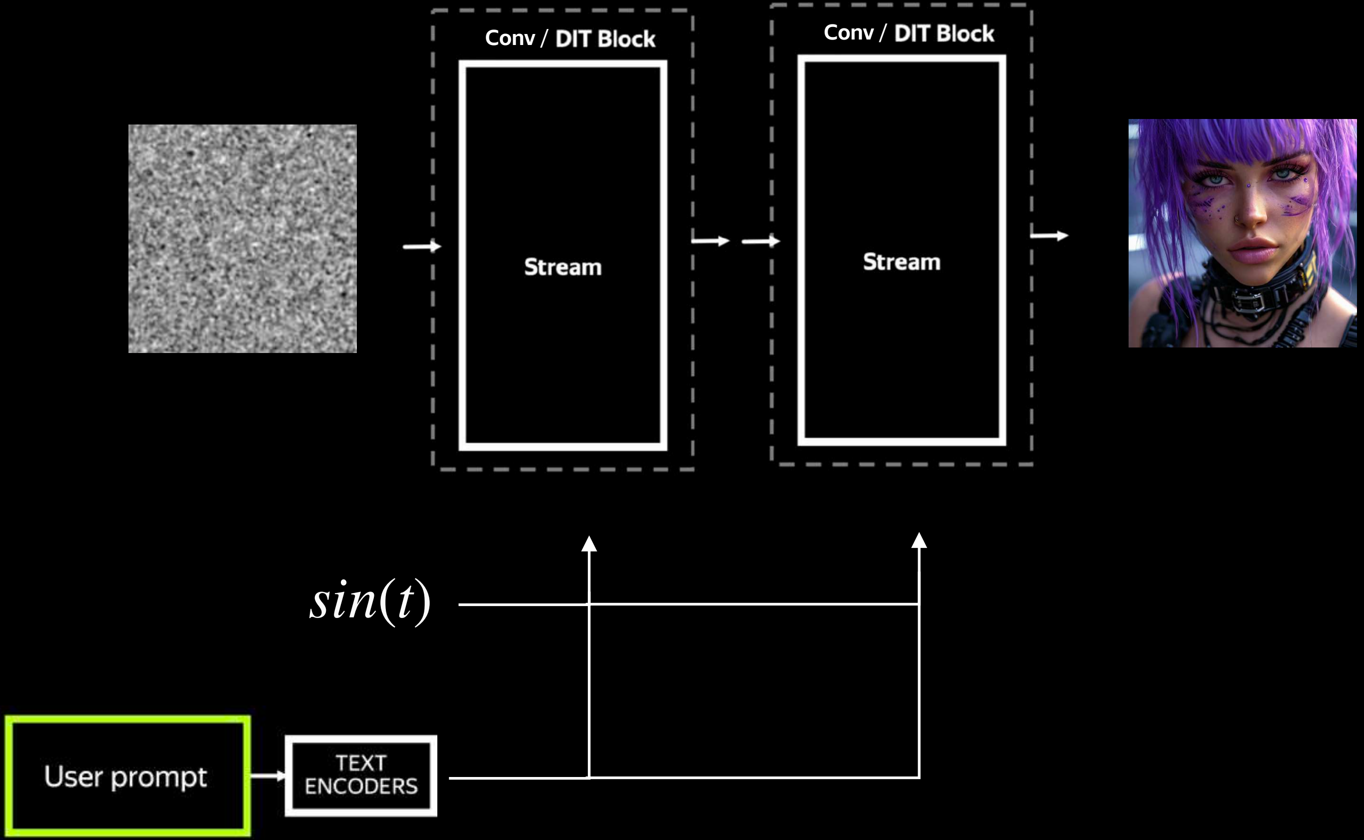
“Cyberpunk girl”

Text-conditional

Basic Conditioning

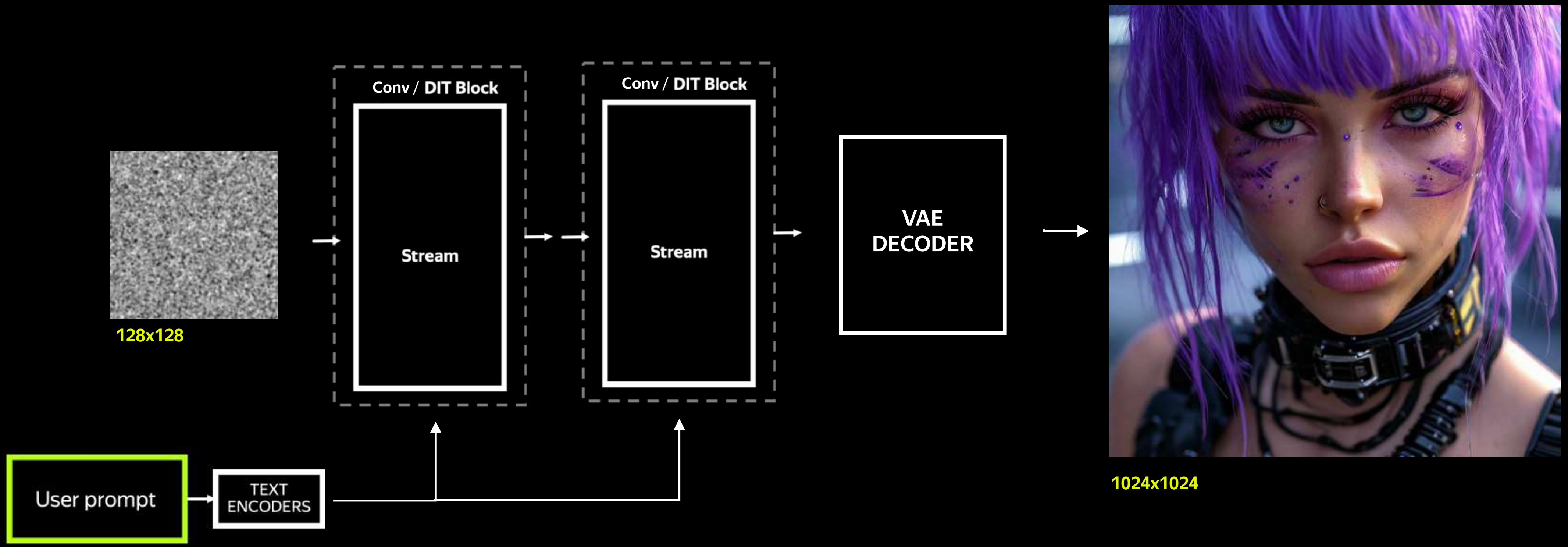
“Cyberpunk girl”

Text-conditional

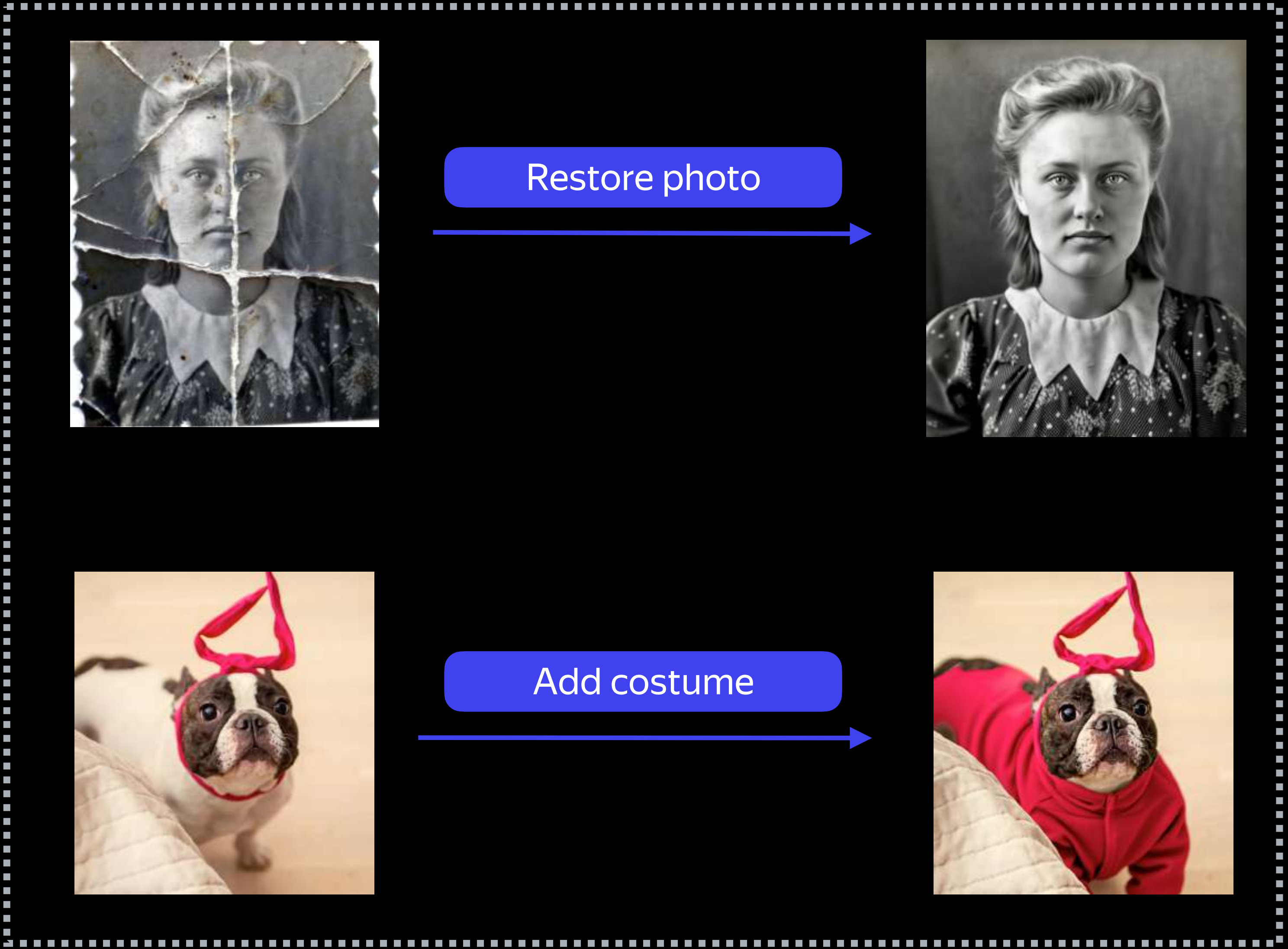


Practical Consideration

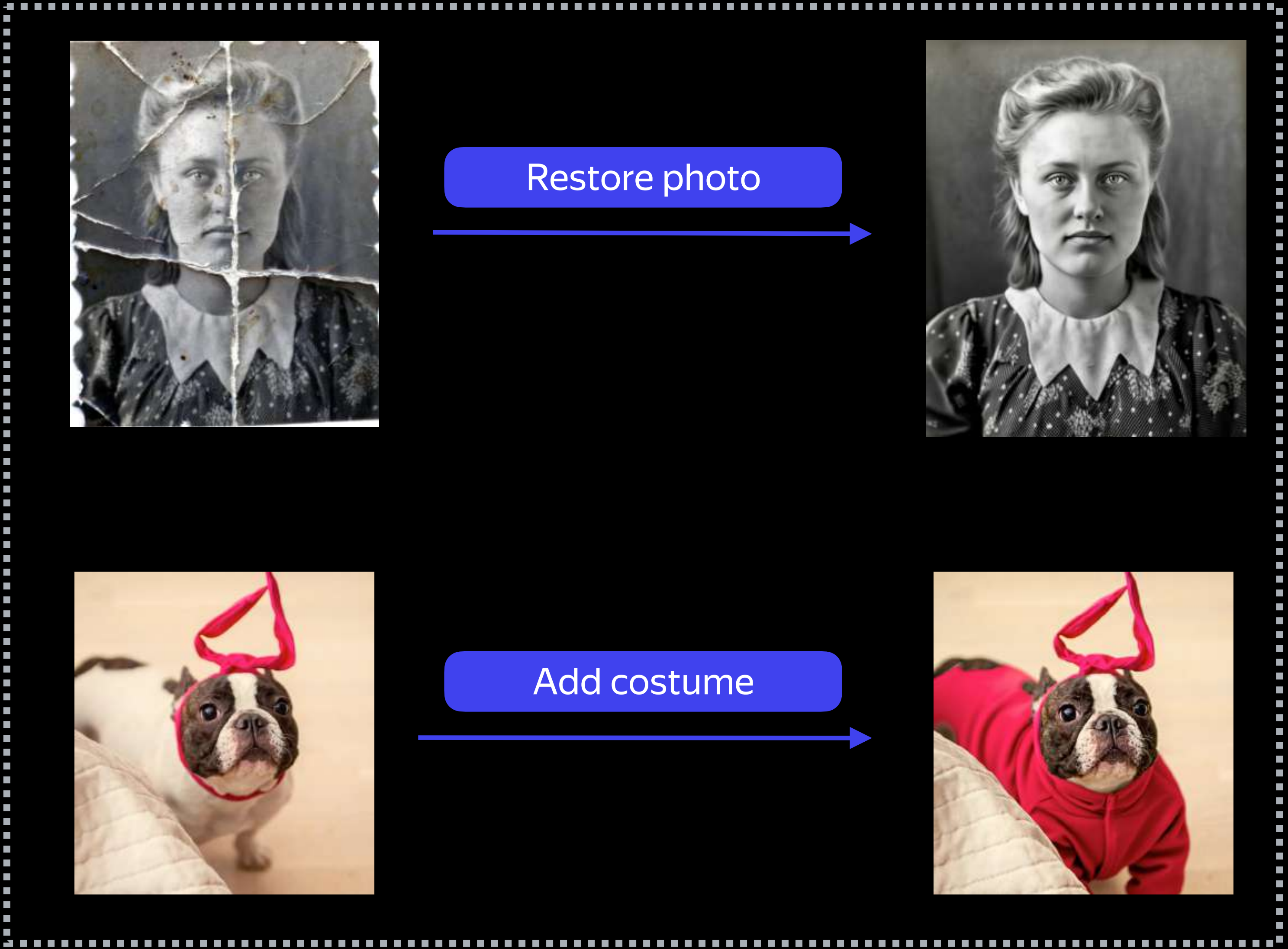
Latent Diffusion



Conditioning on text and image



Conditioning on text and image



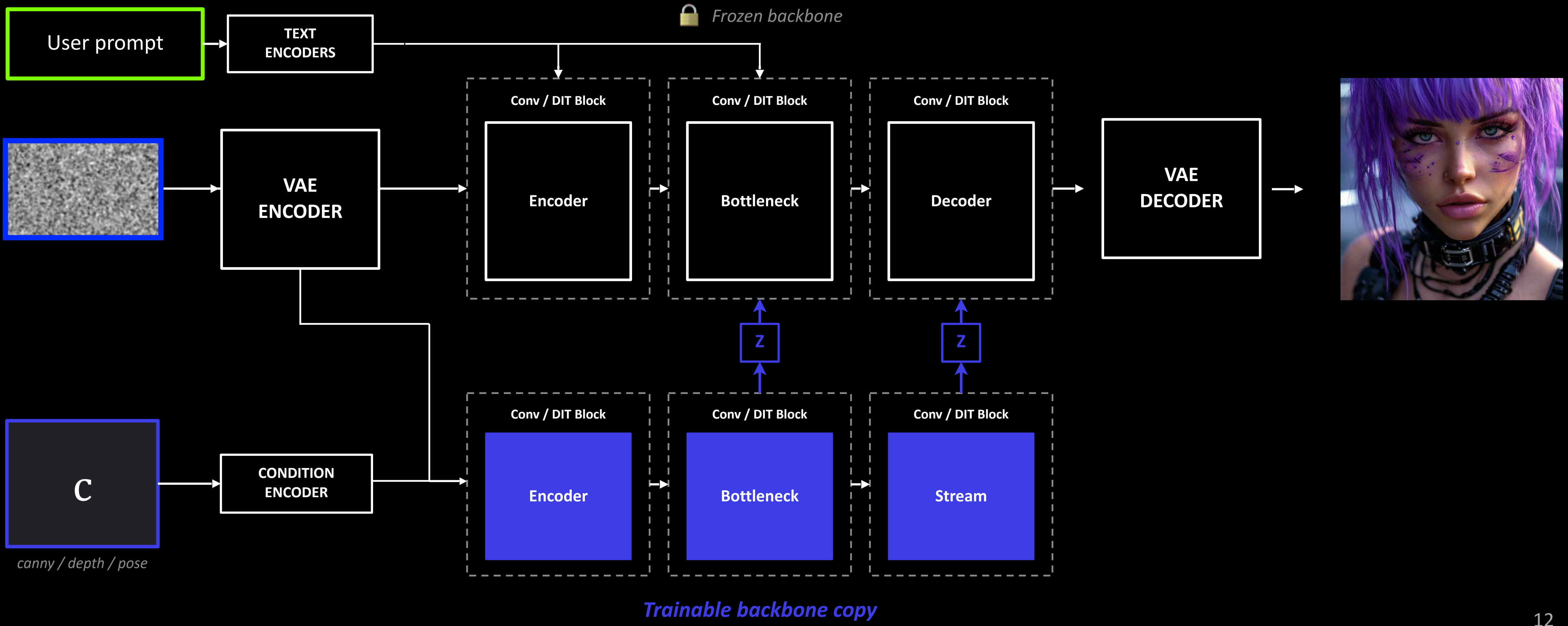
Why is it harder?

Preservation:

- Spatial/structural control
- Semantic/identity control

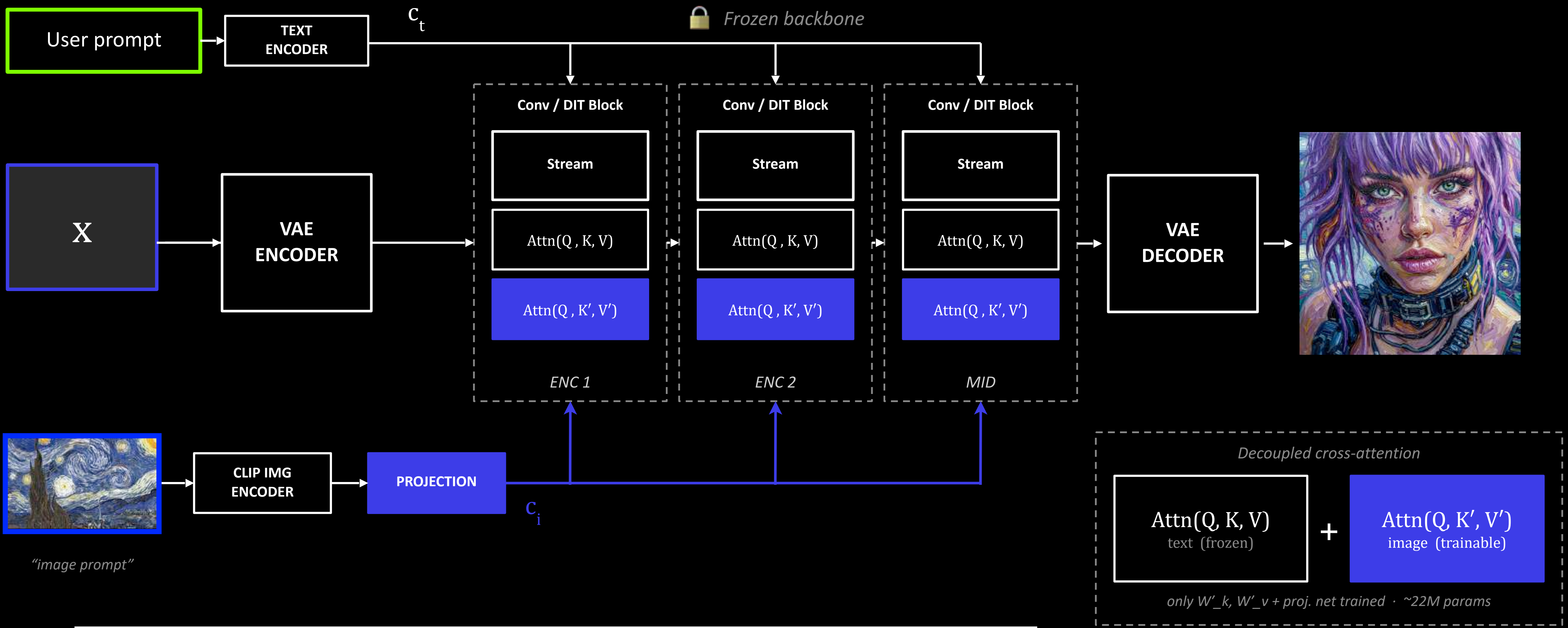
ControlNet

Spatial/Structural Conditioning



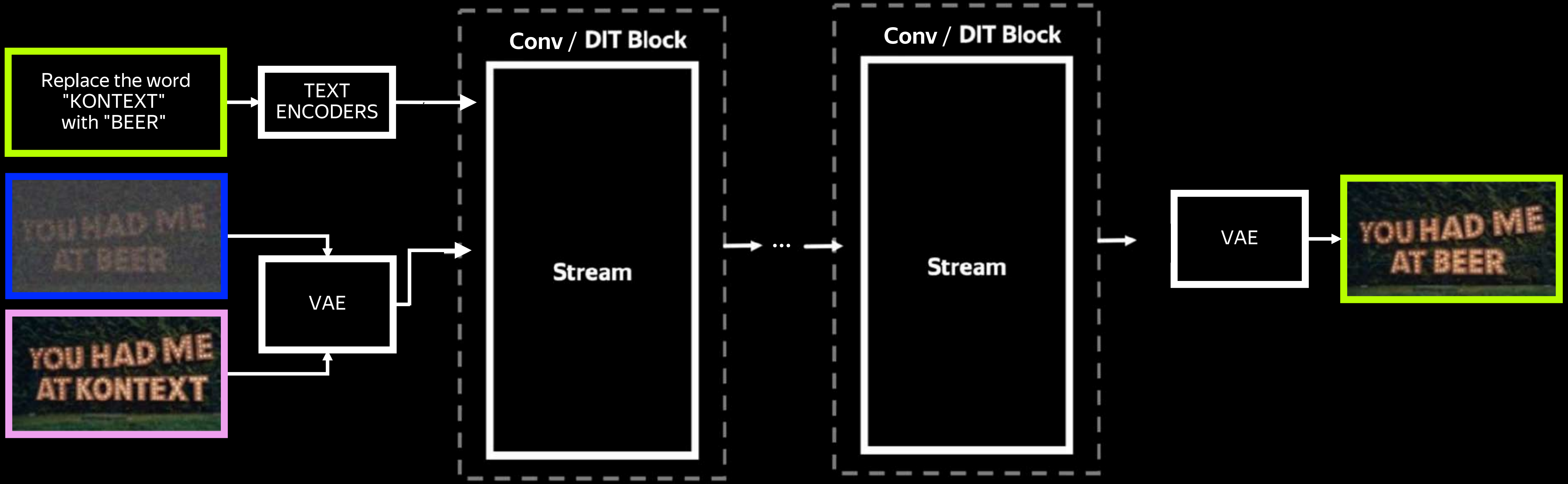
IP-Adapter

Semantic/Identity Conditioning

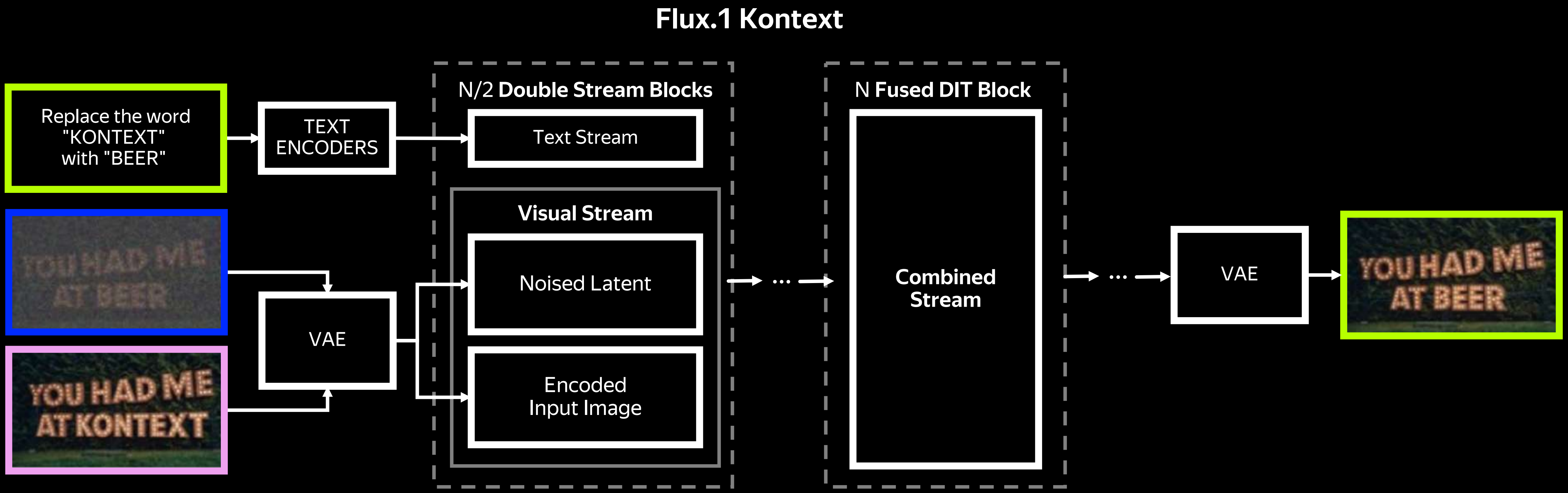


Hu Ye et. al., "IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models", 2023

Transformers - just concat conditions



Transformers - the Flux way



S. Batifol et. al., “FLUX.1 Kontext: Flow Matching for In-Context Image Generation and Editing in Latent Space”, 2025

Classifier-Free Guidance

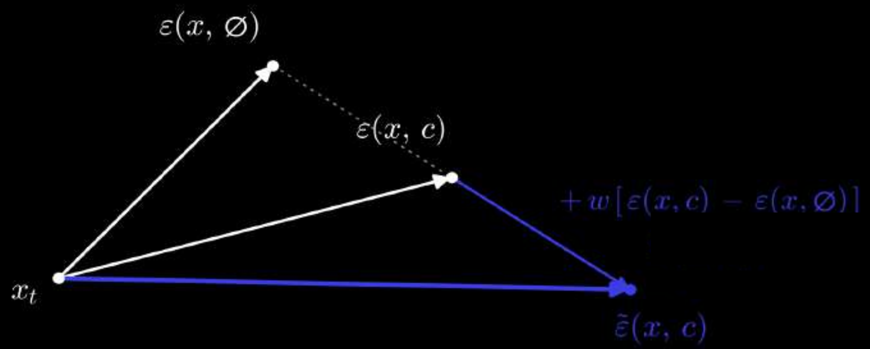
Training

$\mathcal{L} = \mathbb{E}\,\|\varepsilon_\theta(x, c) - \varepsilon\|^2$ $c \leftarrow \varnothing$ with probability p

Inference

$$\tilde{\varepsilon}(x, c) = \varepsilon(x, \varnothing) + (1 + w)\,[\,\varepsilon(x, c) - \varepsilon(x, \varnothing)\,]$$

$$= (1 + w)\cdot\varepsilon(x, c) - w\cdot\varepsilon(x, \varnothing)$$



Extrapolate from $\varepsilon(x, \varnothing)$ through $\varepsilon(x, c)$ by w

Implicit classifier

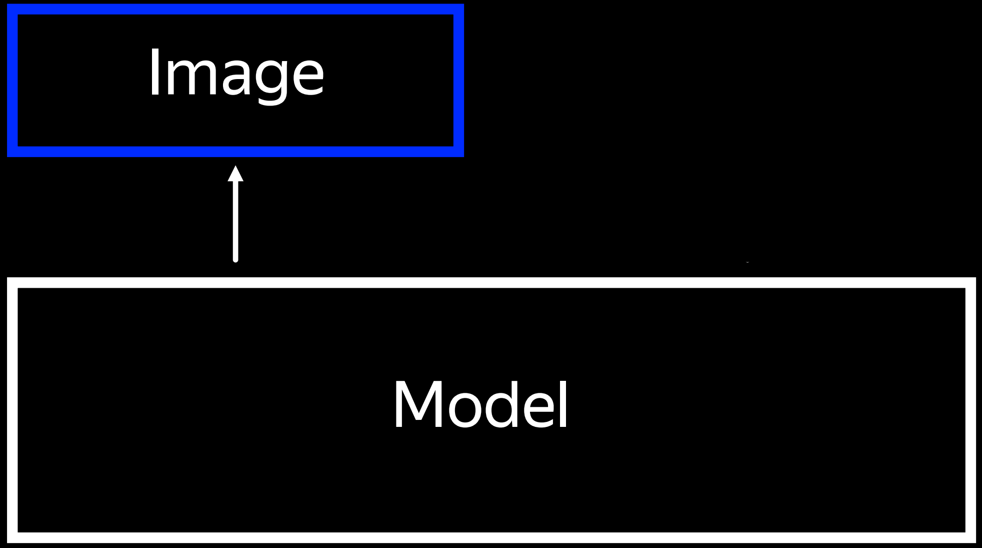
$$\nabla_x \log p(c \mid x) \propto \varepsilon(x, c) - \varepsilon(x, \varnothing)$$

$w = 0$ recovers $\tilde{\varepsilon}(x, c)$

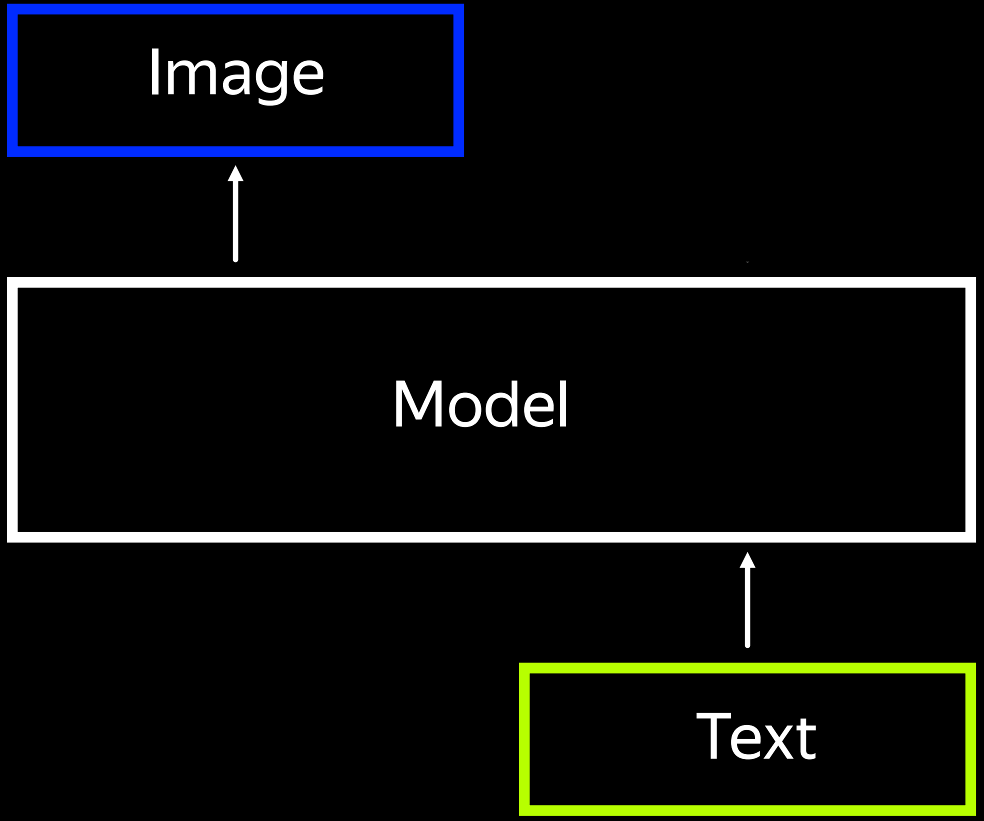
$w \nearrow$ samples from $p(x|c)\cdot p(c|x)^w$ → fidelity ↑, diversity ↓

$w \gg 1$ extrapolation leaves the data manifold

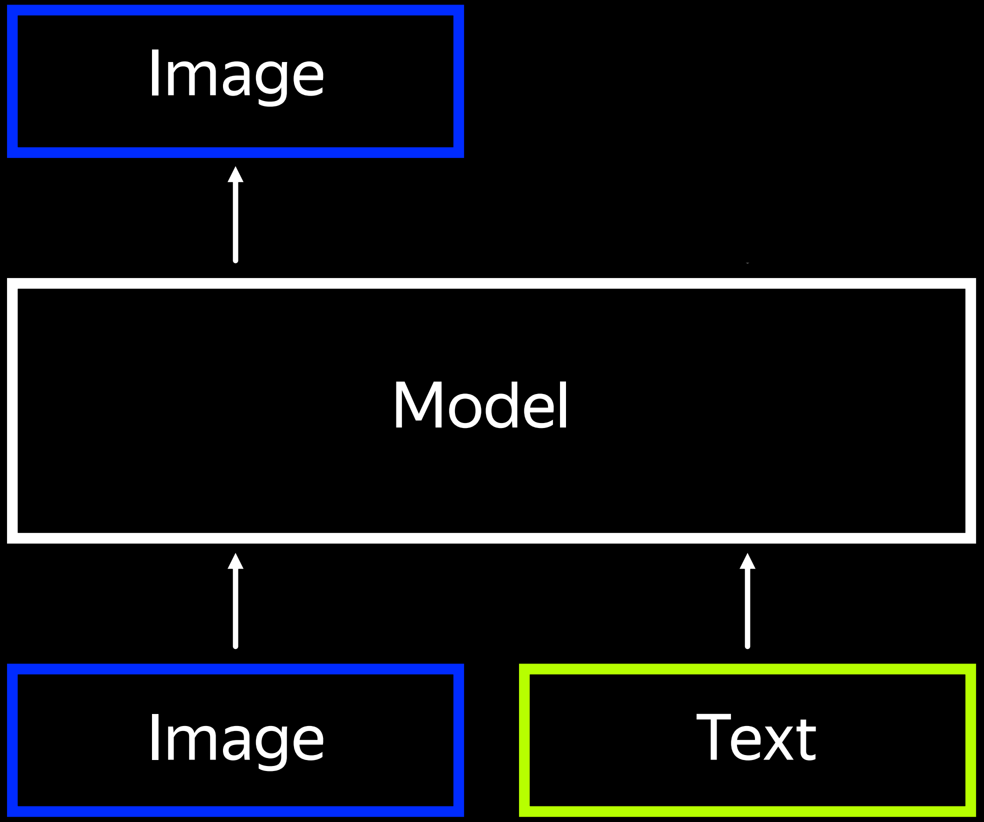
Recap



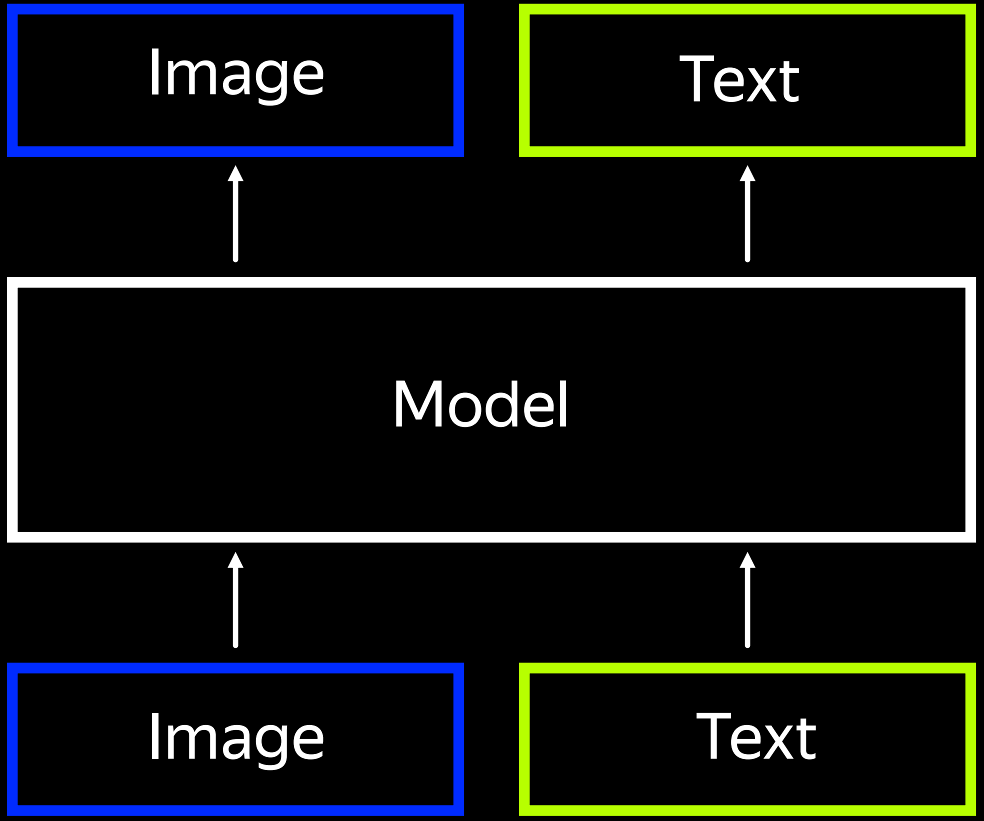
Recap



Recap



Multi-modal Generation



In reality we want multi-modal

Why?

01 Bidirectional capability*

02 Interleaved / any-to-any generation

03 Emergent multimodal reasoning

* J. Zhang et. al., “Are Unified Vision-Language Models Necessary: Generalization Across Understanding and Generation”, 2025

The central design tension

Unify two main generative worlds

- Inherently continuous (visual): images, video, 3D, audio
- Inherently discrete (textual): text, code, math

The central design tension

Unify two main generative worlds

- Inherently continuous (visual): images, video, 3D, audio
- Inherently discrete (textual): text, code, math

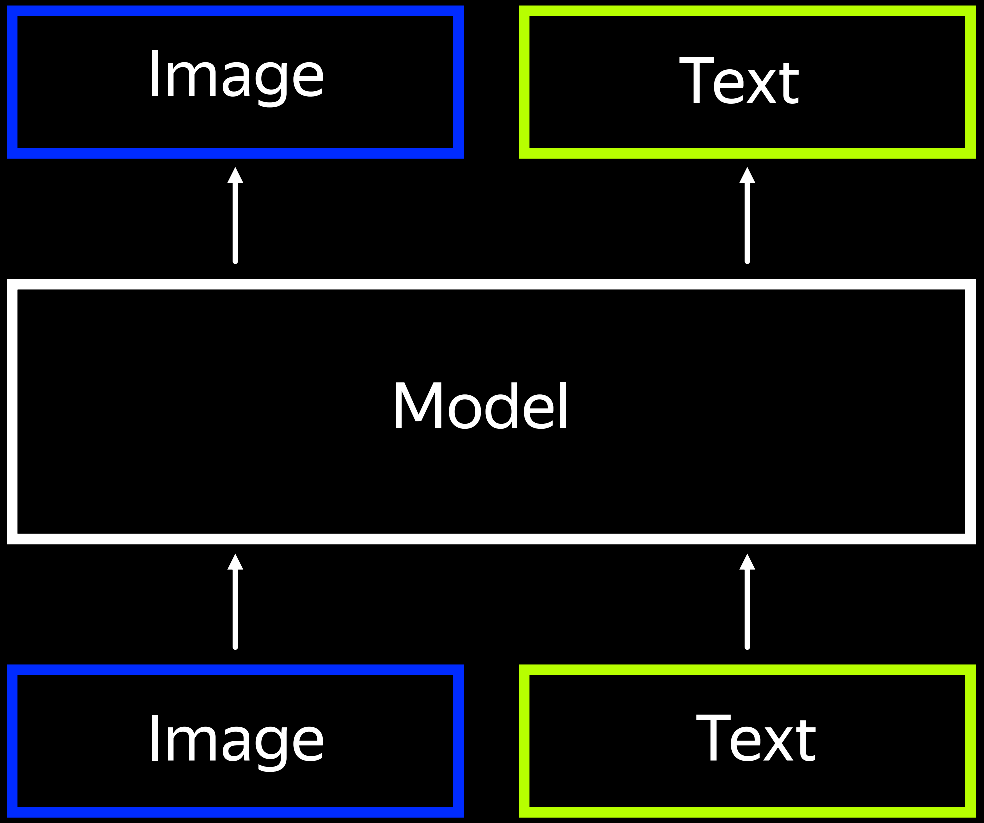
Currently SOTA approaches

- discrete-to-discrete — LLM-like (e.g. aLLM)
- discrete-to-continuous — diffusion models
- mixed-to-discrete — VLM-like (e.g. AVLM)
- mixed-to-continuous — diffusion models

Yandex Research

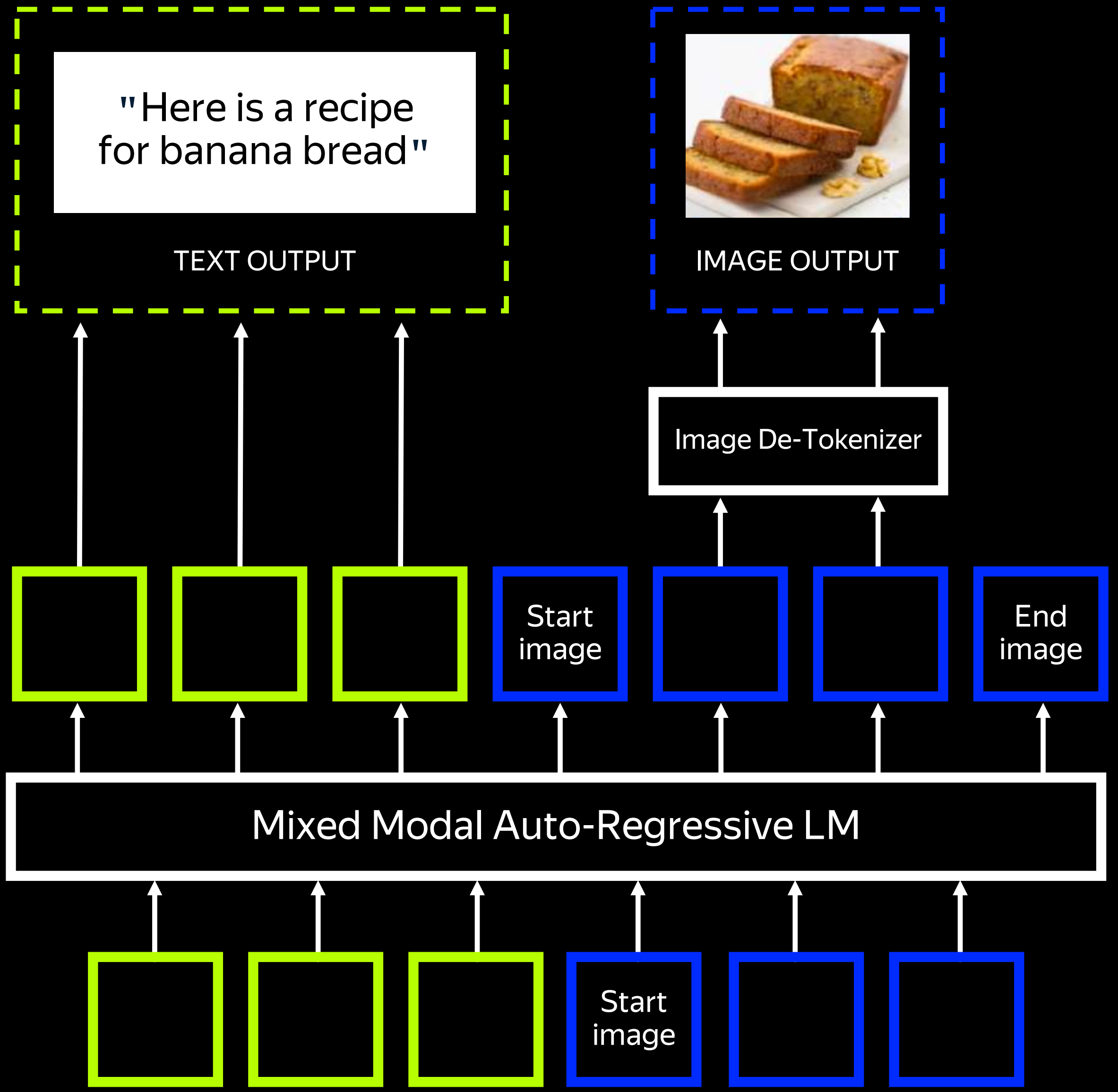
Towards Unified Generative Models

What kind of model would help us?



Single backbone
Single objective

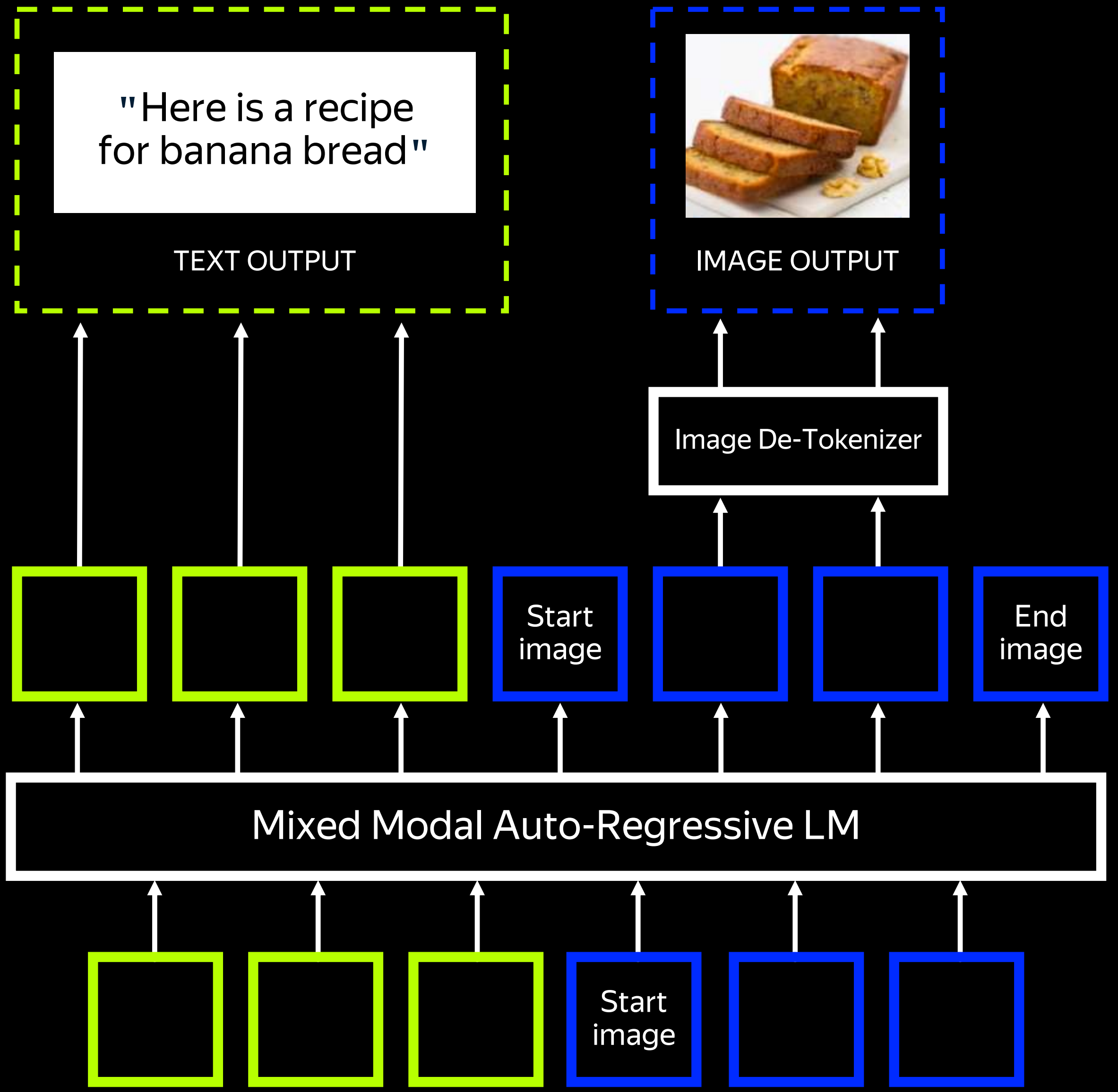
Pure auto-regressive



Single backbone
Single objective

Pure auto-regressive

- (+) Uniform implementation
- (−) Raster order is not native to images
- (−) Discretisation ruins image quality



Single backbone
Single objective

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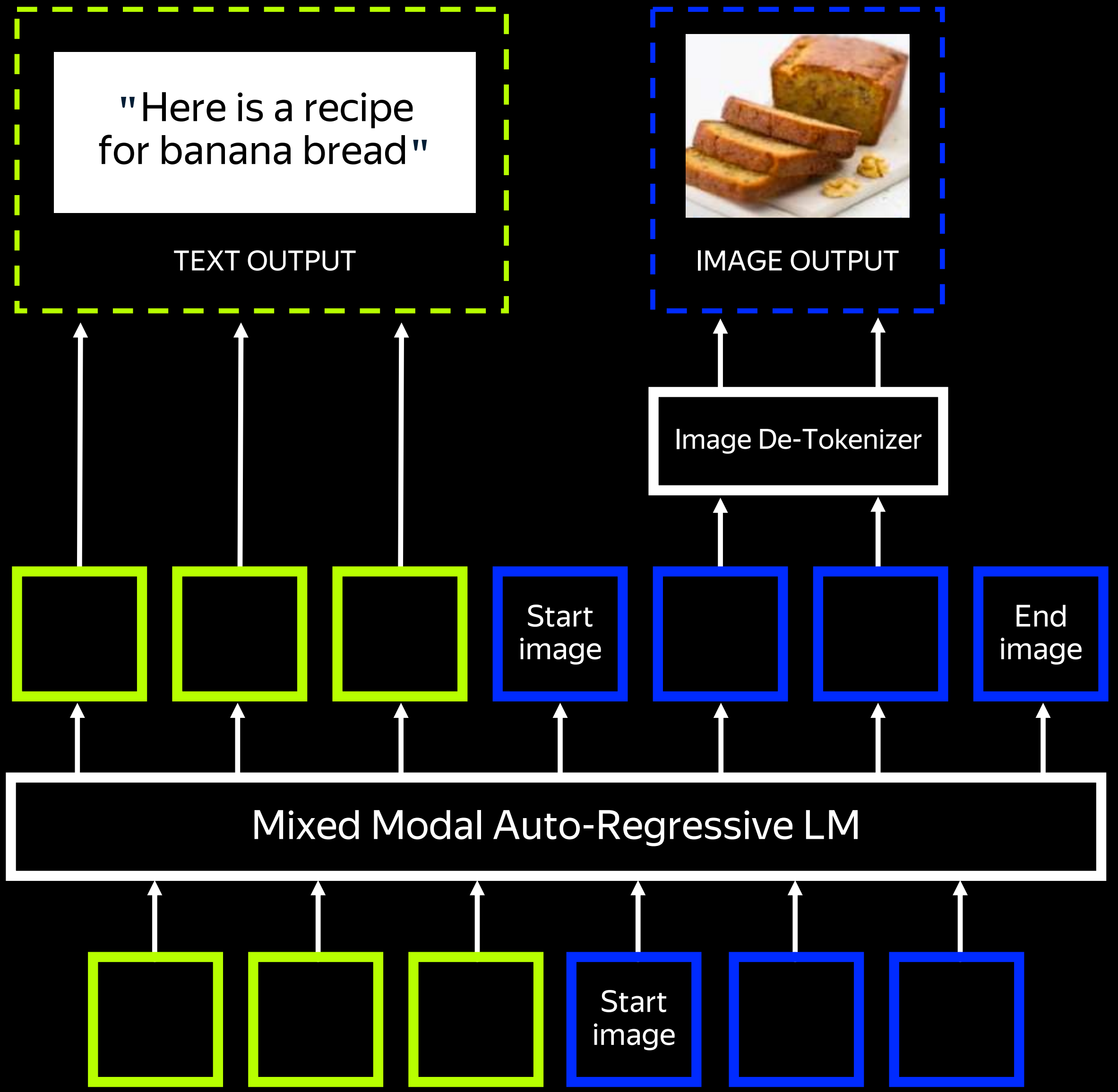
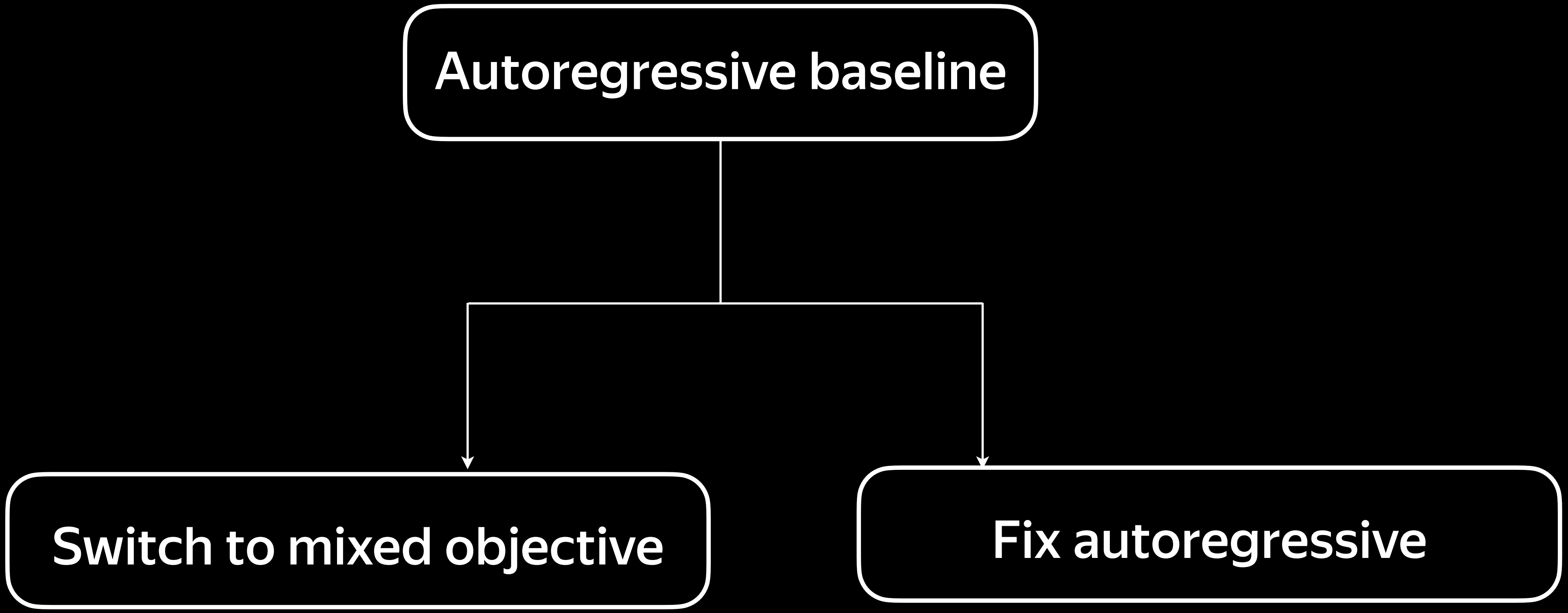
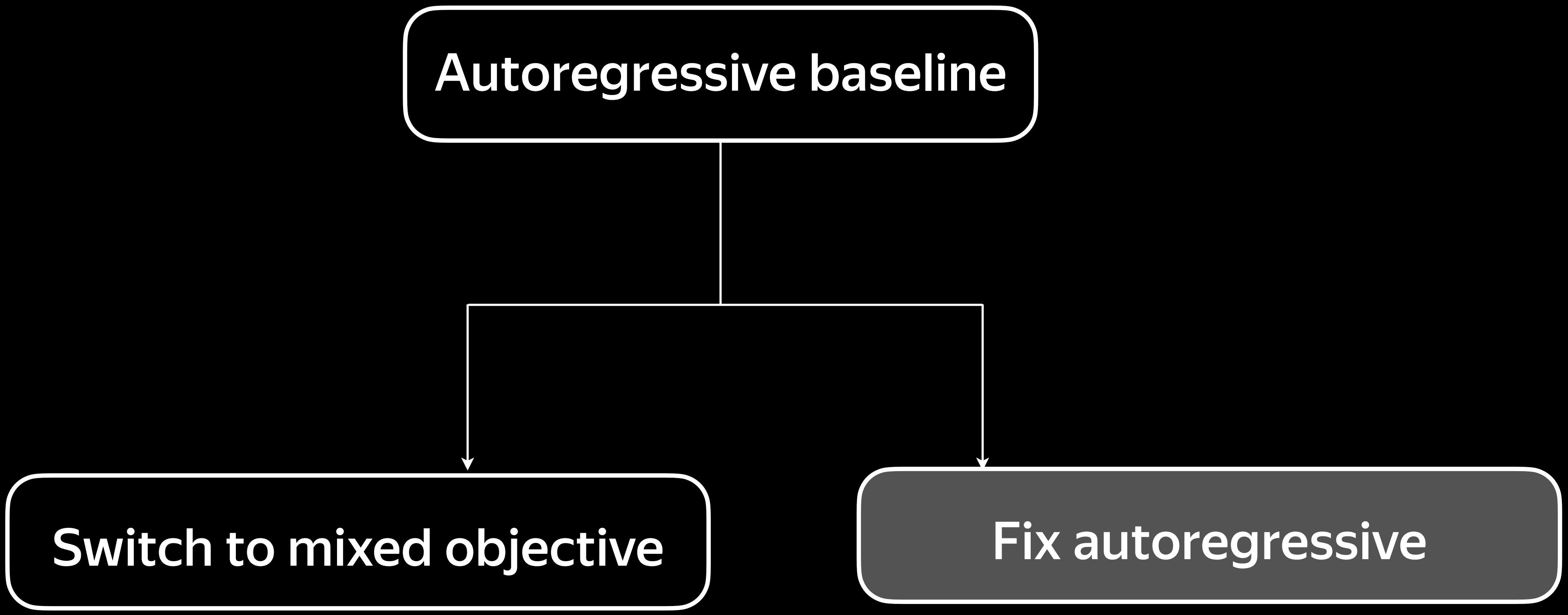
sic!



A. Ramesh et. al., "Zero-Shot Text-to-Image Generation" (aka DALL-E 1), 2021


Autoregressive baseline
Switch to mixed objective
Fix autoregressive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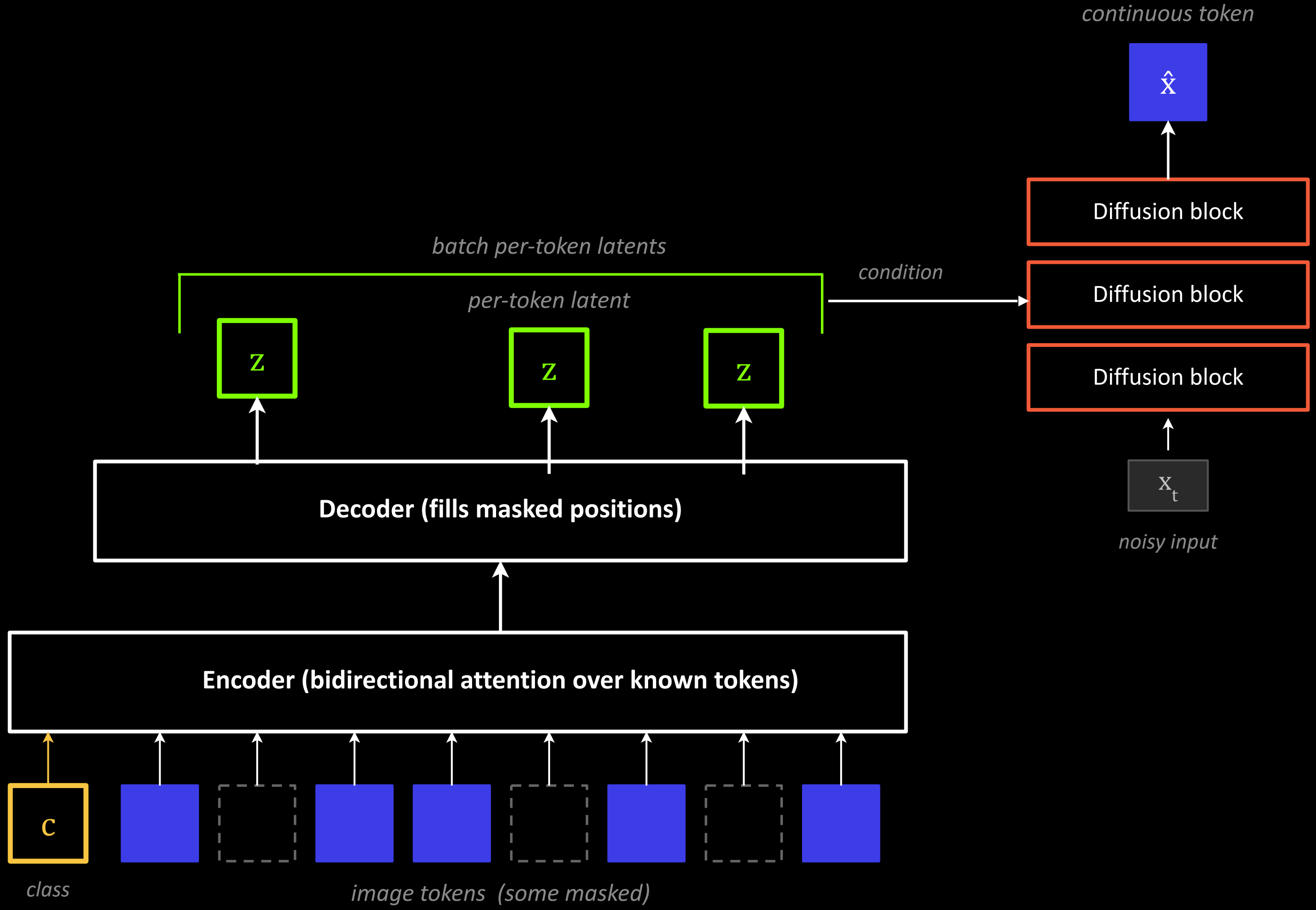
MAR: AR without Vector Quantization

Autoregression over continuous tokens via a per-token diffusion head

+ Continuous tokens via small diffusion head
+ Random-order generation, many tokens per step
+ Bidirectional attention inside the image
− Class-conditioned only (no text prompts yet)

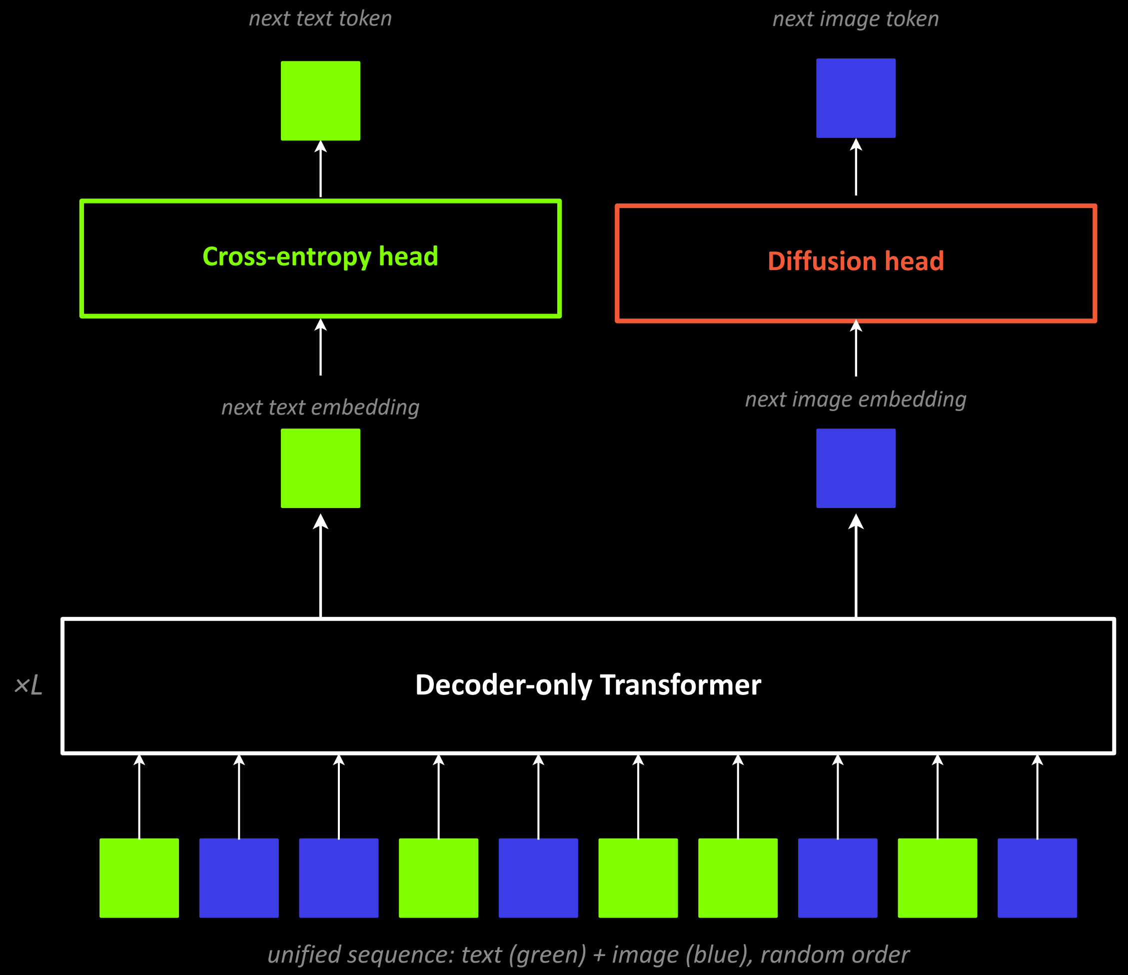


Li et al., "Autoregressive Image Generation without Vector Quantization", NeurIPS 2024

Fluid | UniFluid

Scaling the MAR paradigm to text-to-image

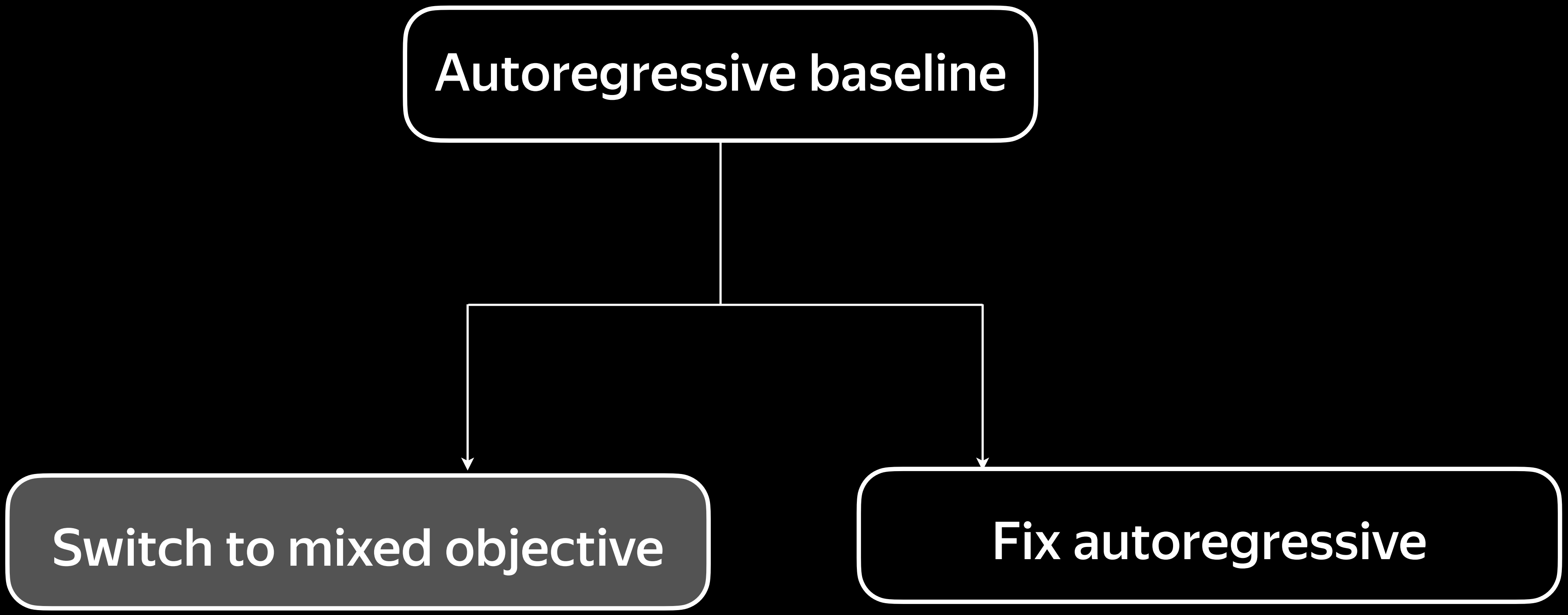
+ Fluid: random-order AR on continuous tokens, scales to 10.5B
+ UniFluid: one decoder-only transformer for gen & understanding
+ Two heads: diffusion head for images, CE head for text
− Tasks trade off — loss balance weight must be tuned carefully



$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda \cdot \mathcal{L}_{\mathrm{diff}}$$

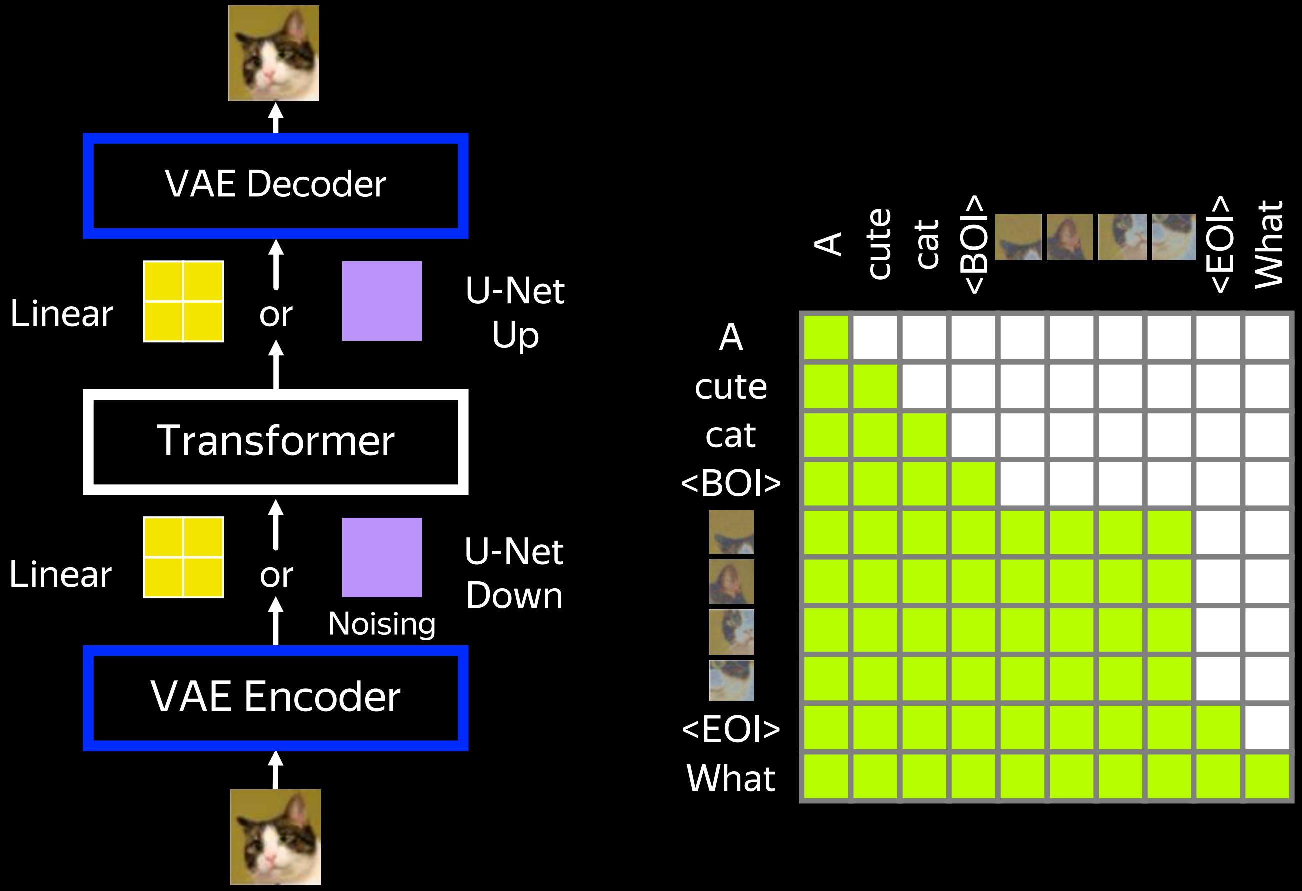
joint next-token loss — balance weight λ matters

Fan et al., Fluid, 2024 · Fan et al., UniFluid, 2025 · Li et al., MAR, 2024


Autoregressive baseline
Switch to mixed objective
Fix autoregressive

Single backbone
Mixed objective

Transfusion



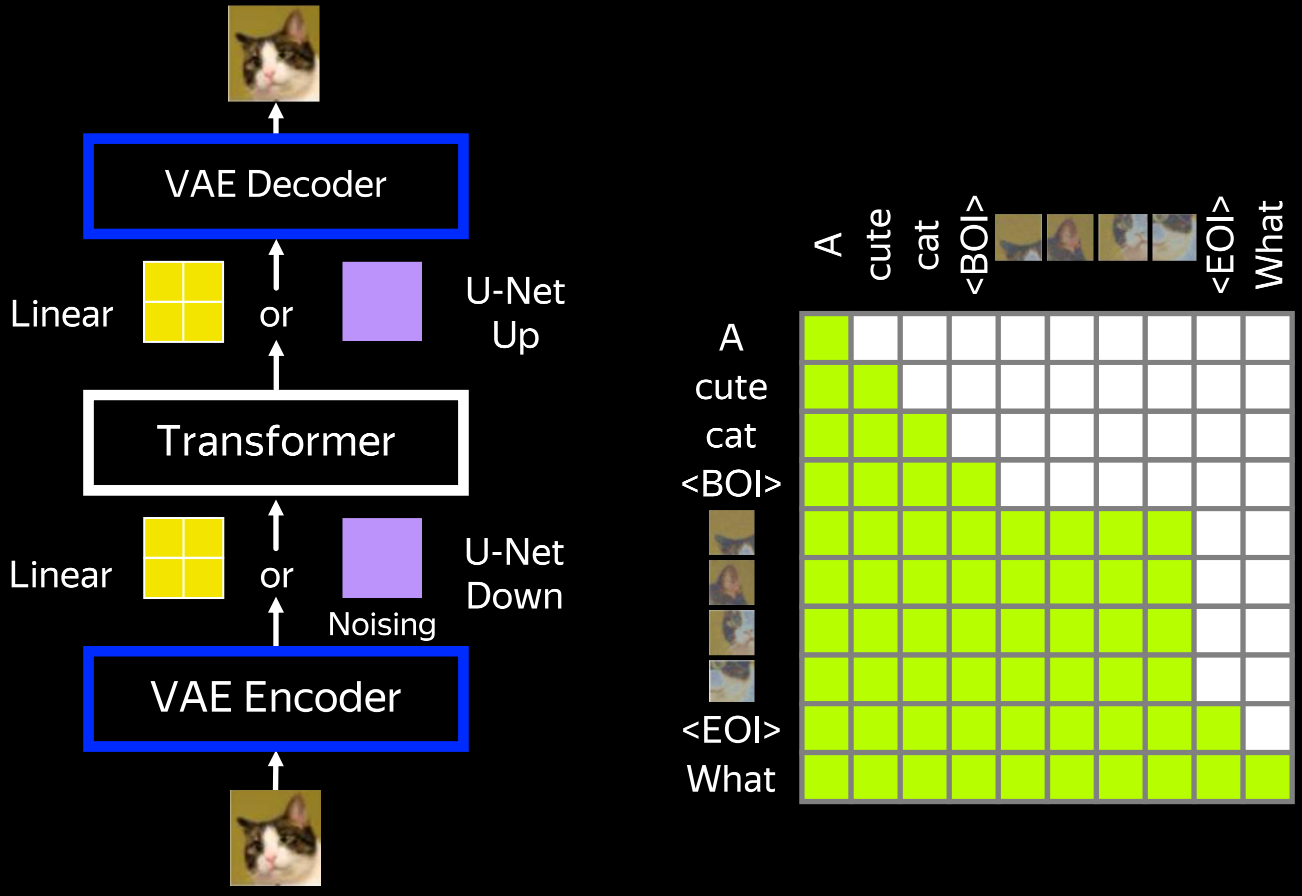
$$\mathcal{L}_{Transfusion} = \mathcal{L}_{LM} + \lambda \times \mathcal{L}_{DDPM}$$

C. Zhou et. al., “Transfusion: Predict the Next Token and Diffuse Images with One Multi-Modal Model”, 2024

Single backbone
Mixed objective

Transfusion

- Unified representation in a single transformer
- Causal attn for text, bidirectional for images
- Continuous encoding of images



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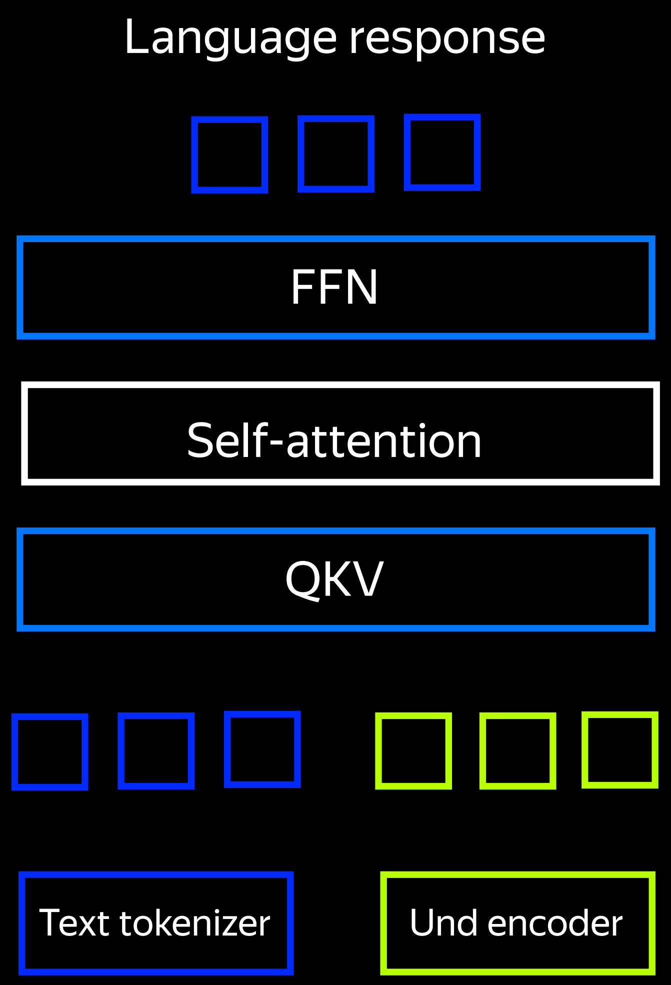
C. Zhou et. al., “Transfusion: Predict the Next Token and Diffuse Images with One Multi-Modal Model”, 2024

Two backbones - Mixed objective

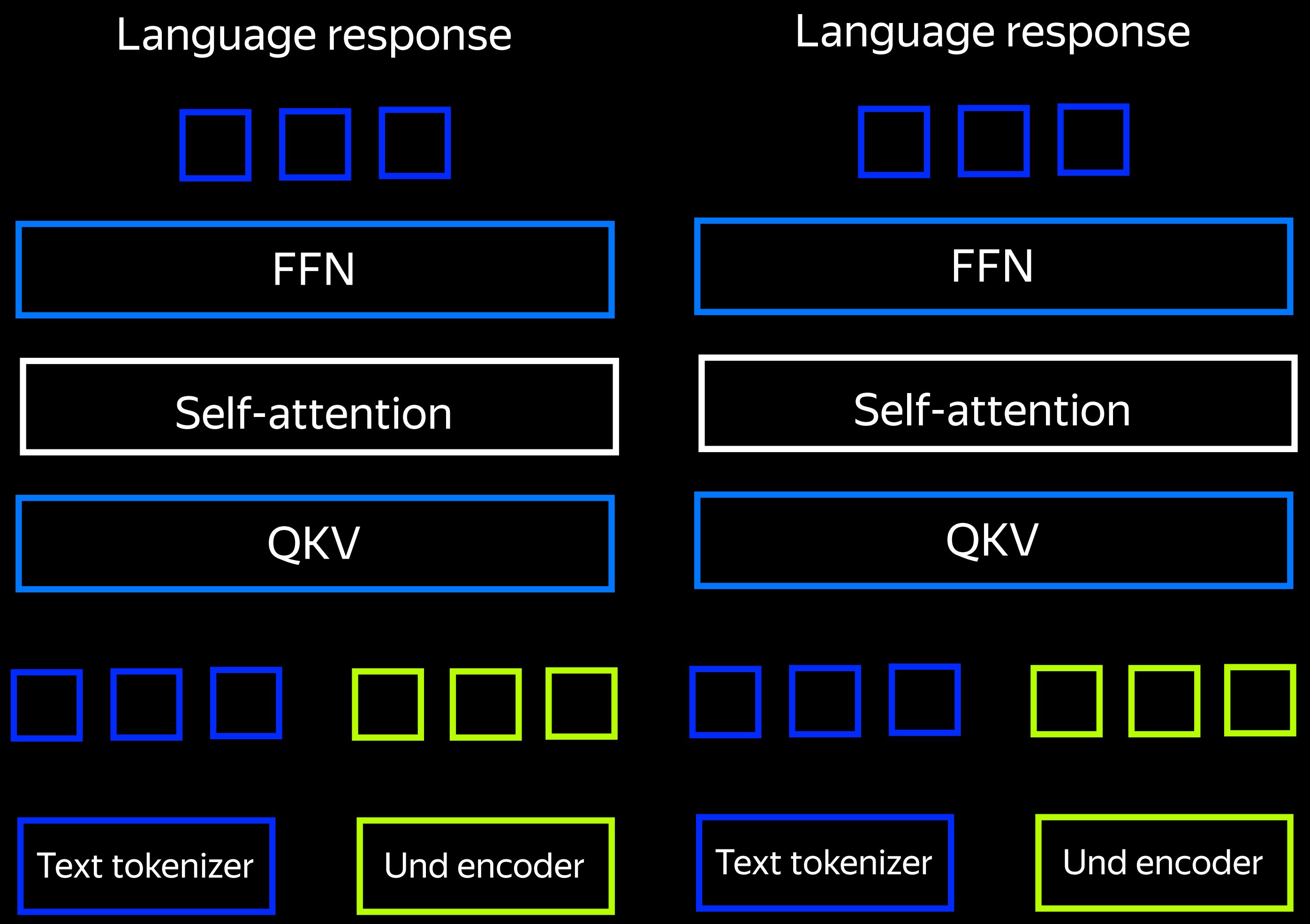
BAGEL



C. Deng et. al., “Emerging Properties in Unified Multimodal Pretraining”, 2025

Two backbones - Mixed objective

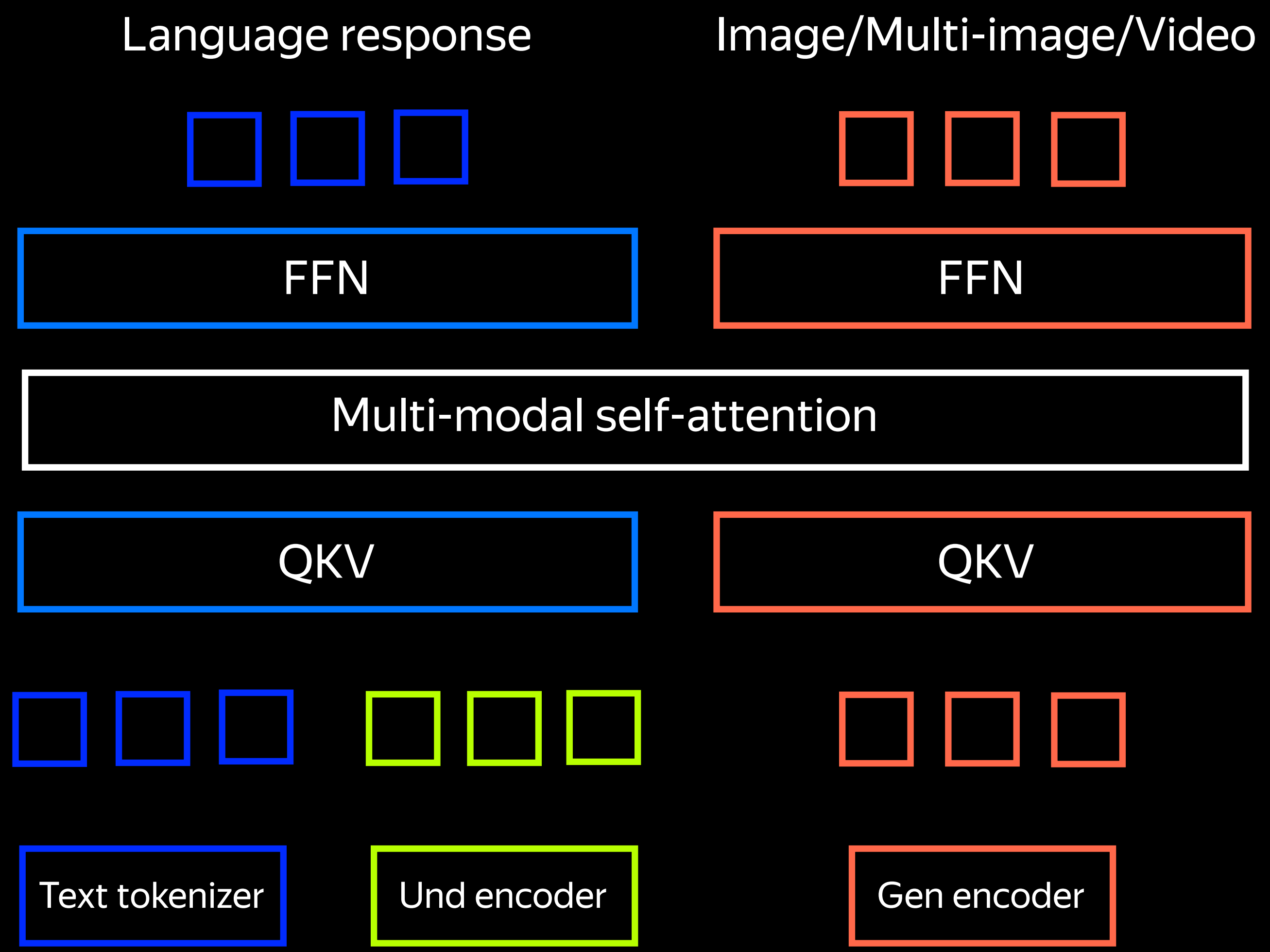
BAGEL



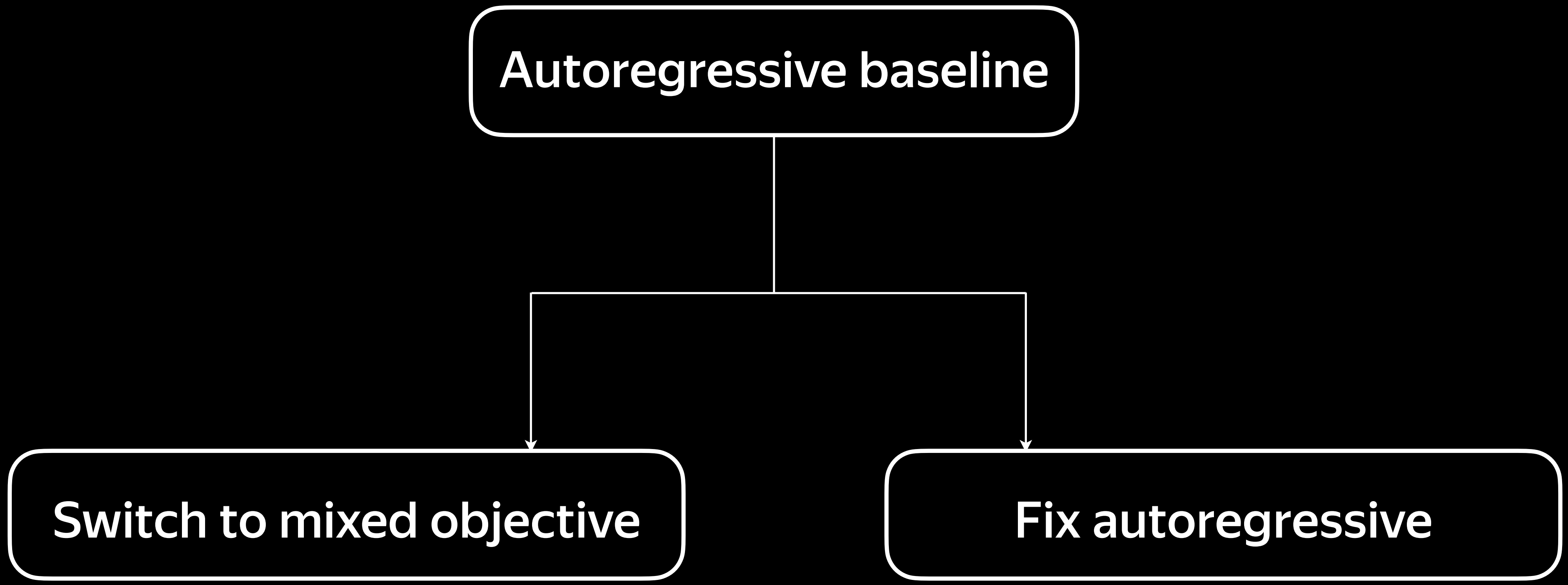
C. Deng et. al., “Emerging Properties in Unified Multimodal Pretraining”, 2025

Two backbones - Mixed objective

BAGEL



C. Deng et. al., “Emerging Properties in Unified Multimodal Pretraining”, 2025


Autoregressive baseline
Switch to mixed objective
Fix autoregressive

Autoregressive baseline

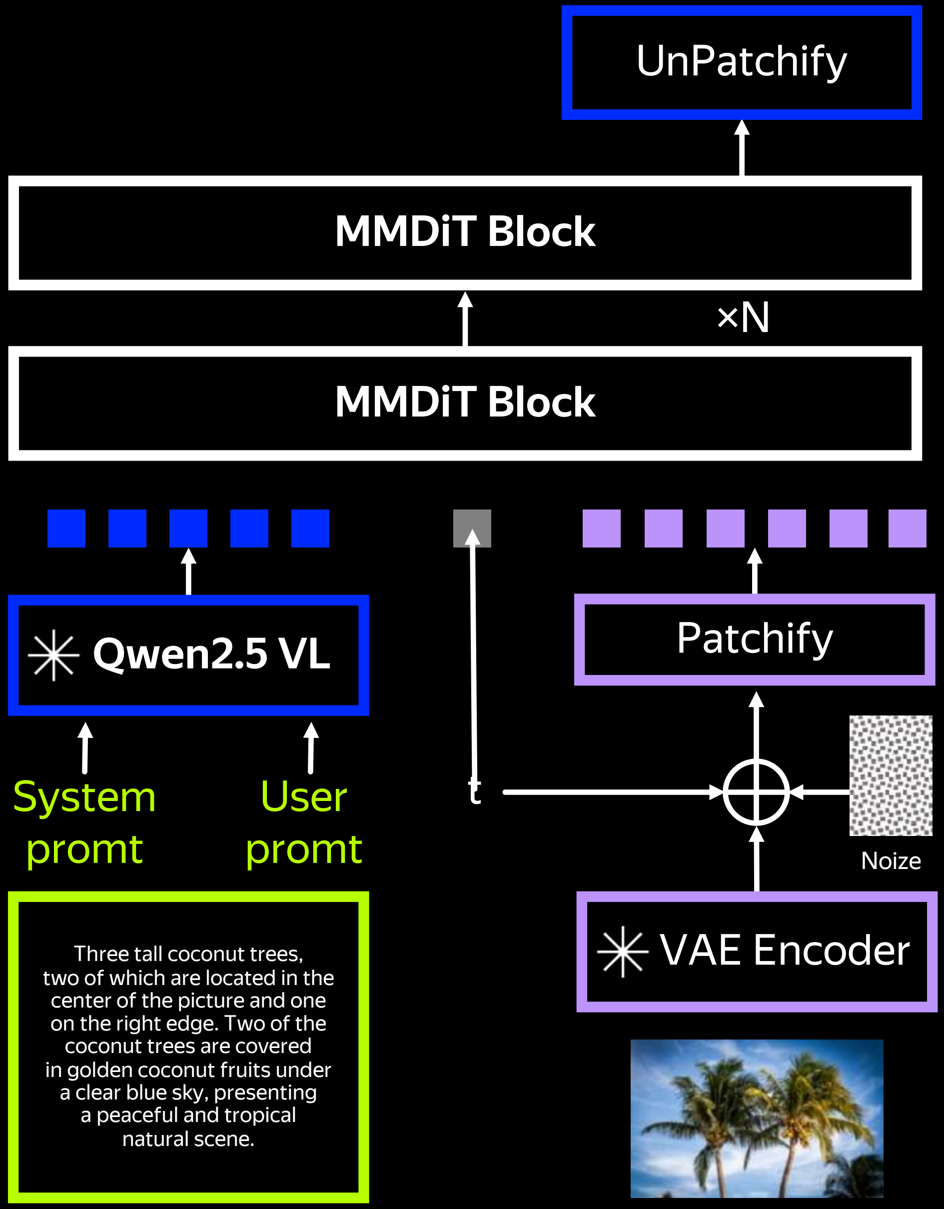
Switch to mixed objective

Fix autoregressive

Just use VLM text encoder

NexusGen | Qwen Image

- (+) Re-use pre-trained diffusion denoiser
- (+) Should be cheaper to train, right?*
- (−) Hard to align with good quality



C. Wu et. al., “Qwen-Image Technical Report”, 2025

C. Wu et. al., “Qwen-Image Technical Report”, 2025

How hard is it to train the connector?

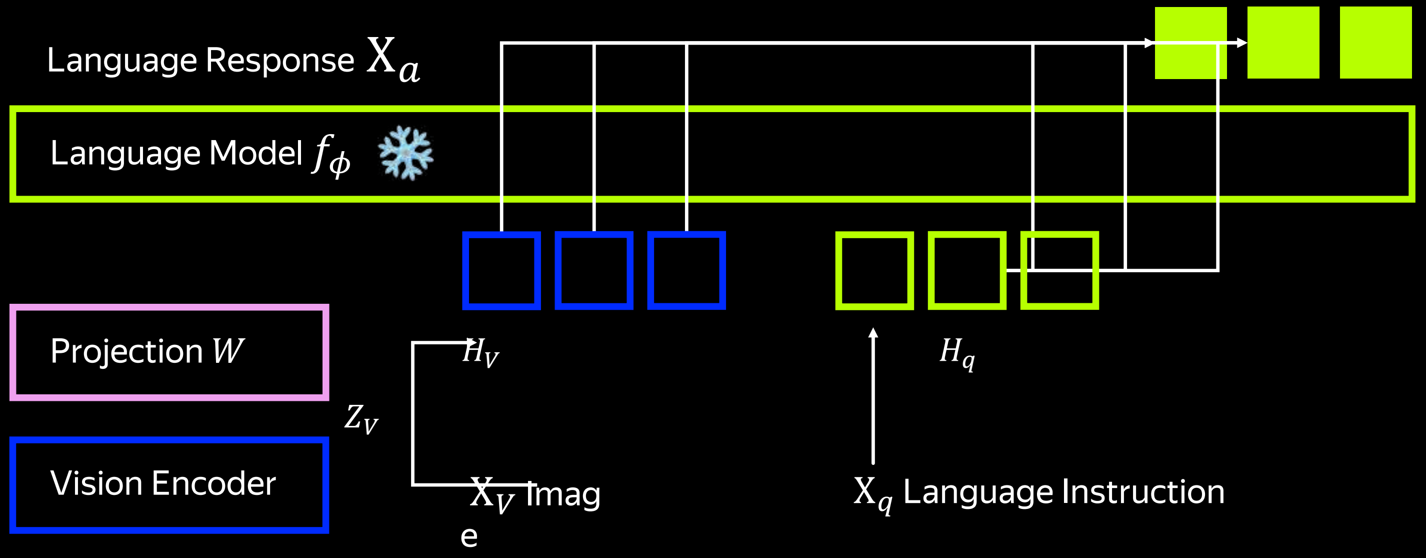
“a pair of cherries, dressed in a delicate ballerina outfit”

- Images generated with **NexusGen** have low quality
- Did not train from scratch

- Images generated with **Qwen Image** are much better
- Trained from scratch

* Very much a hand-waving argumentation, lots of other variables are not aligned between setups

Baseline connection scheme



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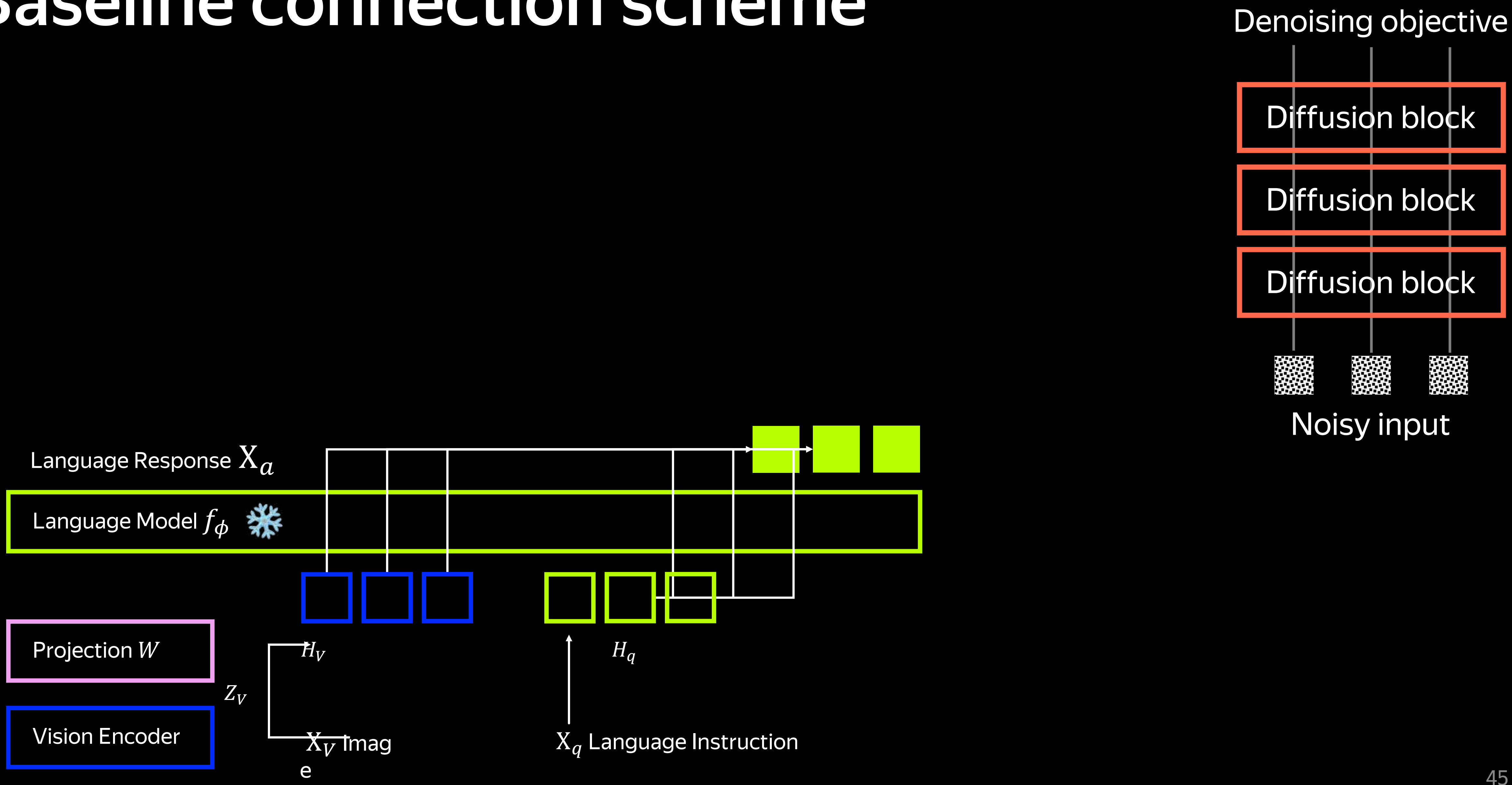
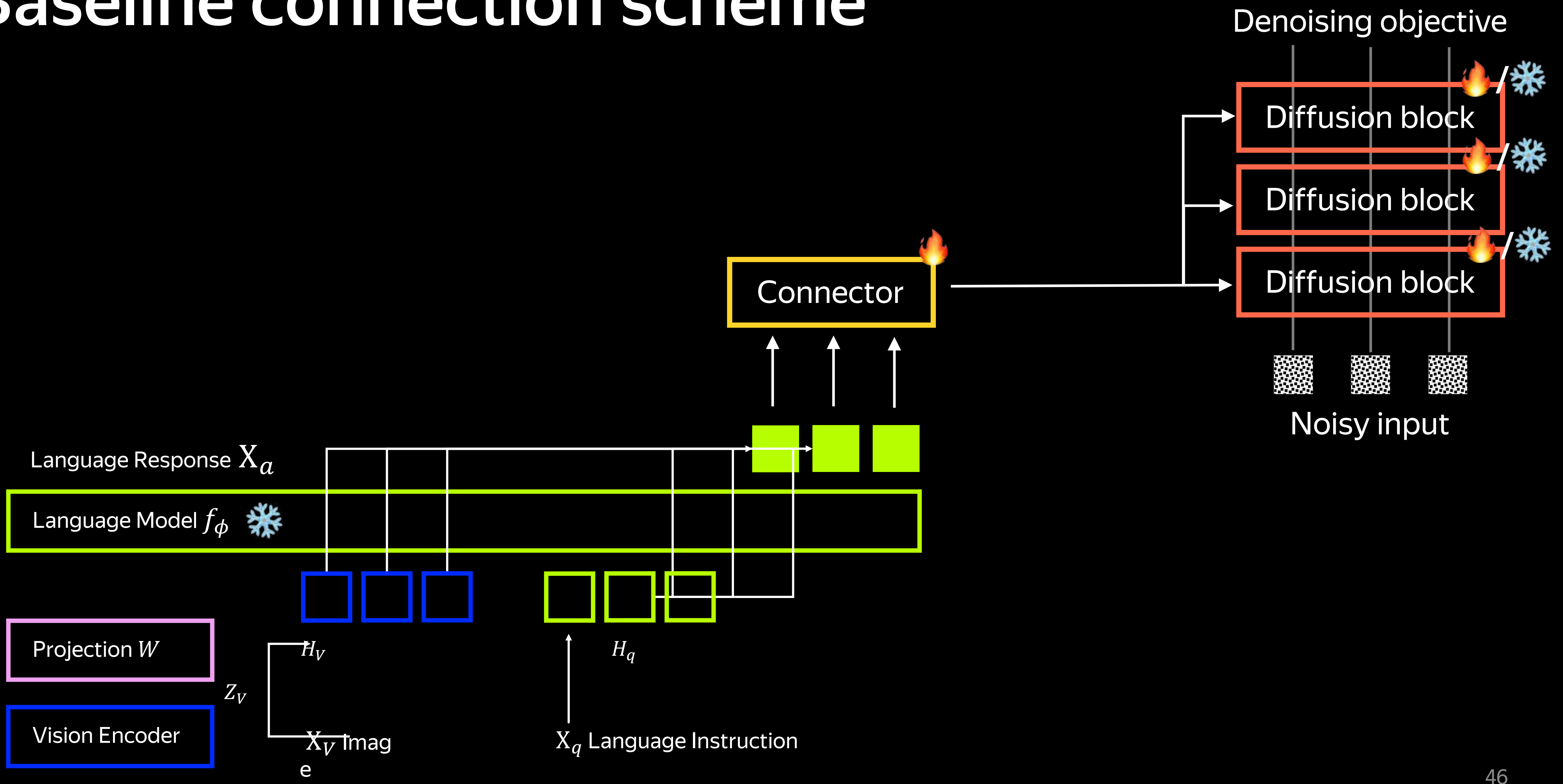
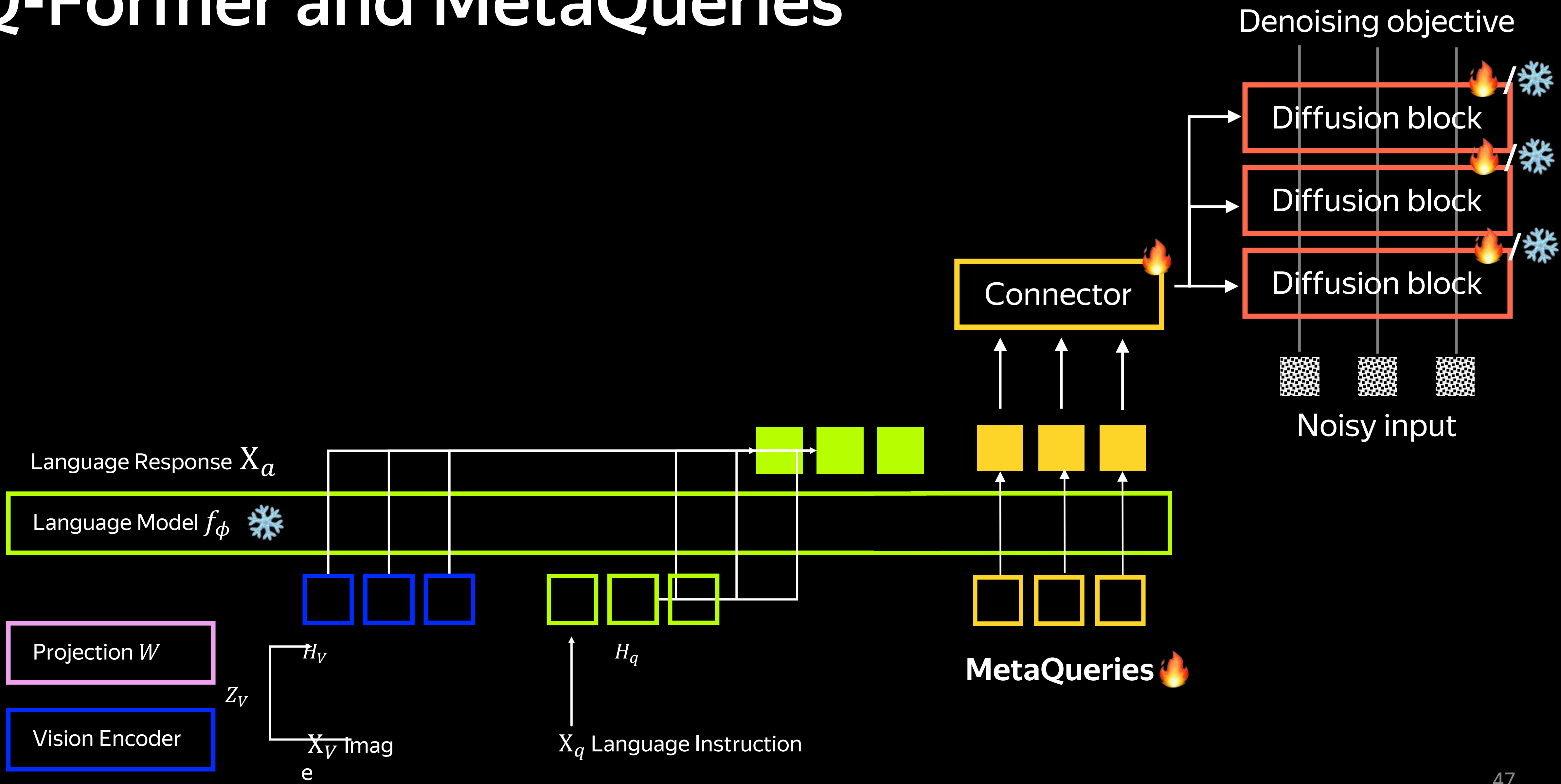


Baseline connection scheme

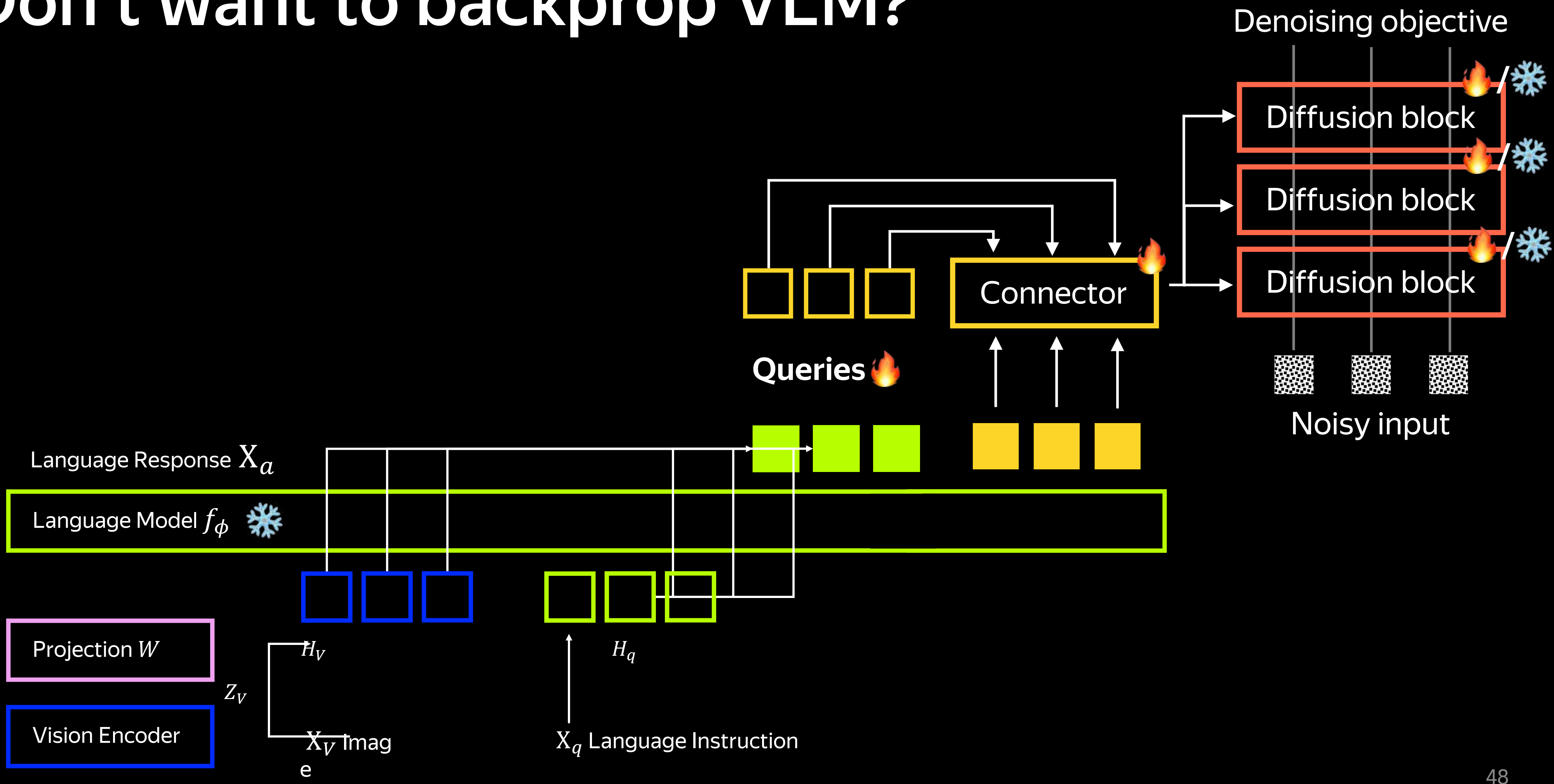


Q-Former and MetaQueries

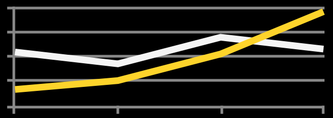


X. Pan et. al., “Transfer between Modalities with MetaQueries”, 2025

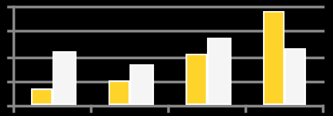
Don’t want to backprop VLM?



Recap

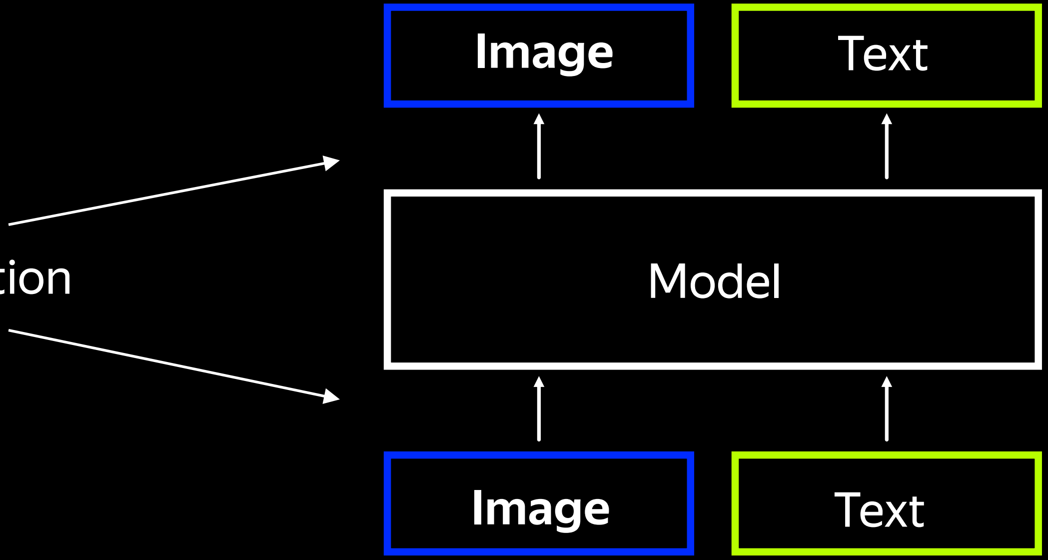


Continuous



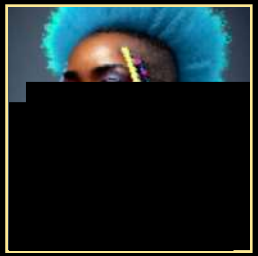
Discrete

Images representation



Recap

Diffusion



Autoregression

Loss

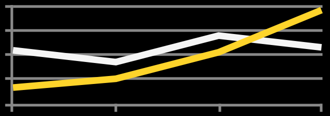
Image

Text

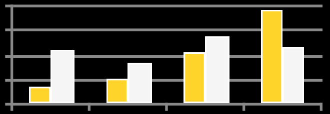
Model

Image

Text



Continuous

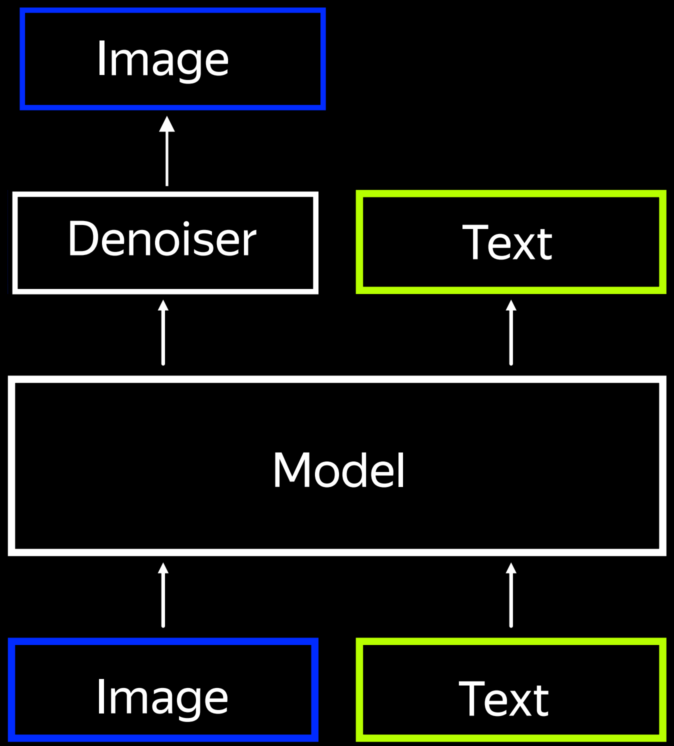


Discrete

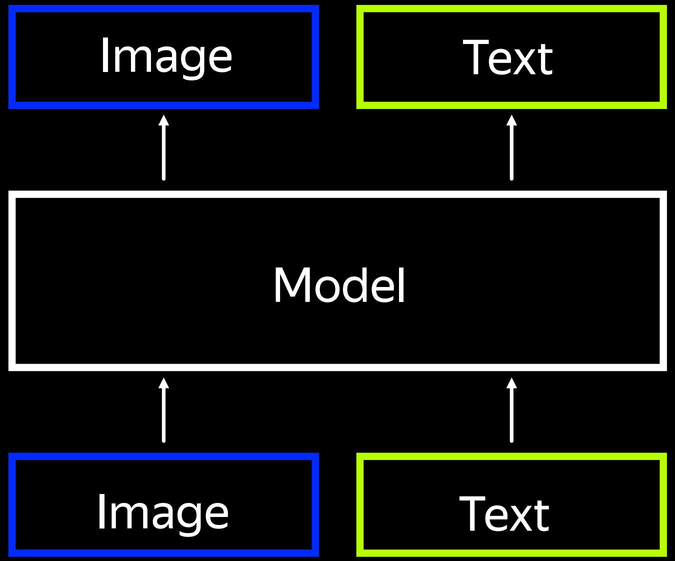
Images representation

Recap

Image gen module



External



Integrated

Recap

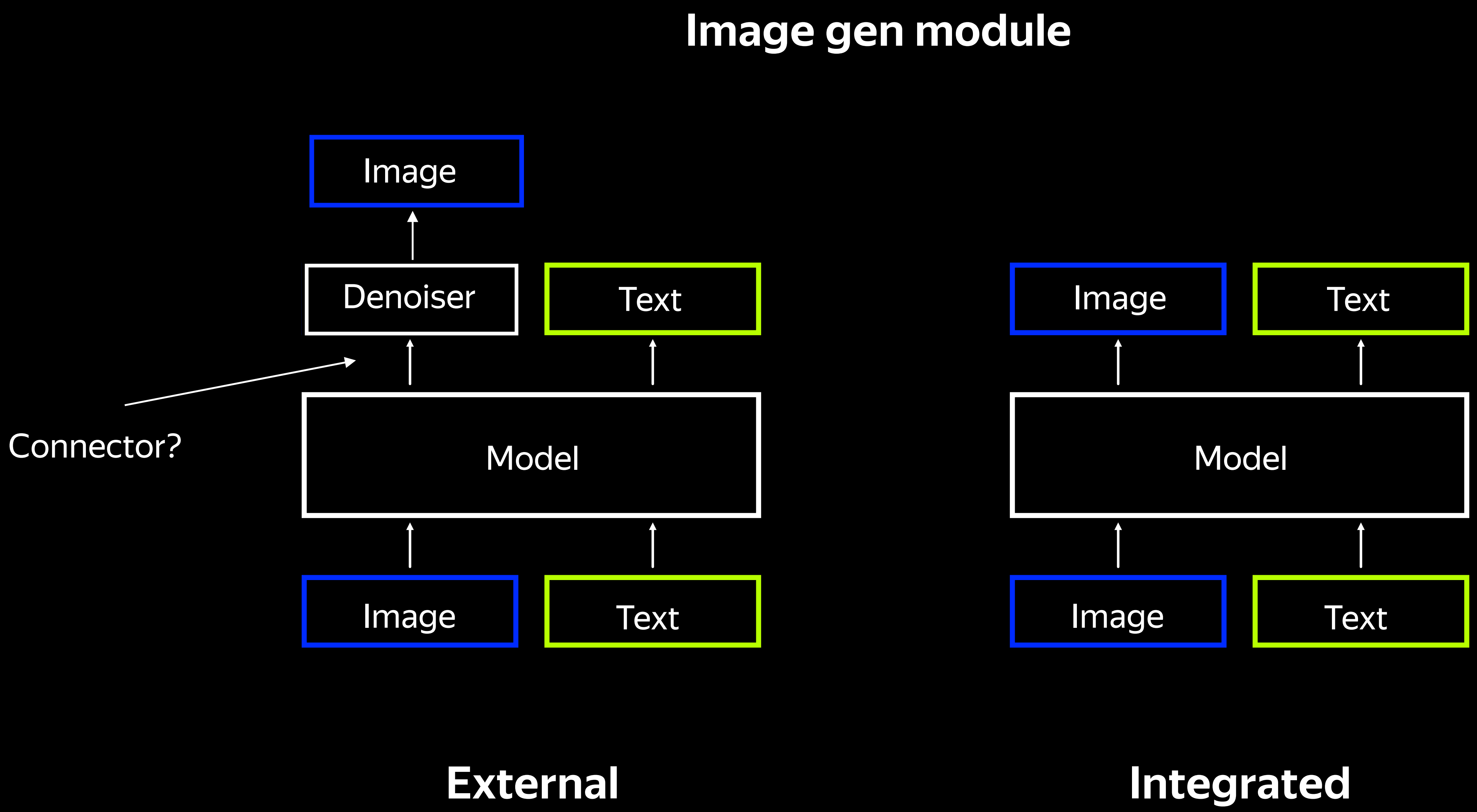


Image gen module
Image
Denoiser
Text
Connector?
Model
Image
Text
Image
Text
Model
Image
Text
External
Integrated

Yandex Research

Inference-time Compute Scaling

Very much known for LLMs

Question

↓

"Reasoning"
Large Language Model

↓

Thinking w/text...

Answer!

Very much known for LLMs

Question

↓

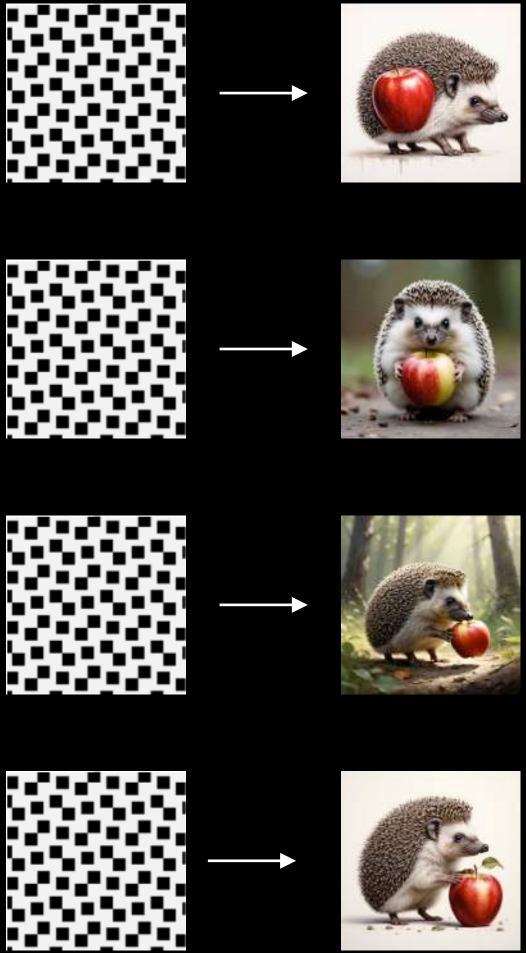
"Reasoning"
Diffusion/Multi-modal model

↓

Thinking w/text+images...

Answer!

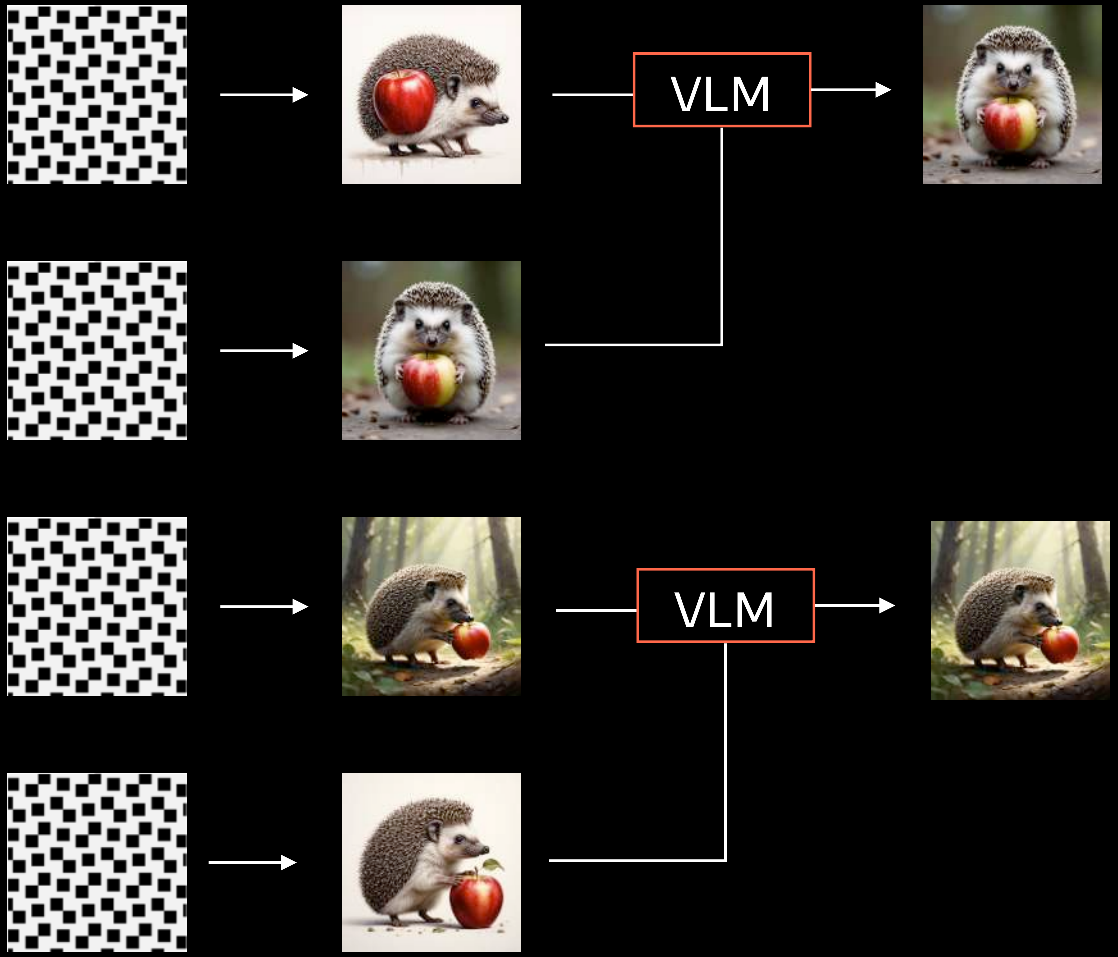
Best-on-N baseline



“hedgehog with an apple”

E. Xie et. al., “SANA 1.5: Efficient Scaling of Training-Time and Inference-Time Compute in Linear Diffusion Transformer“, 2025

Best-on-N baseline



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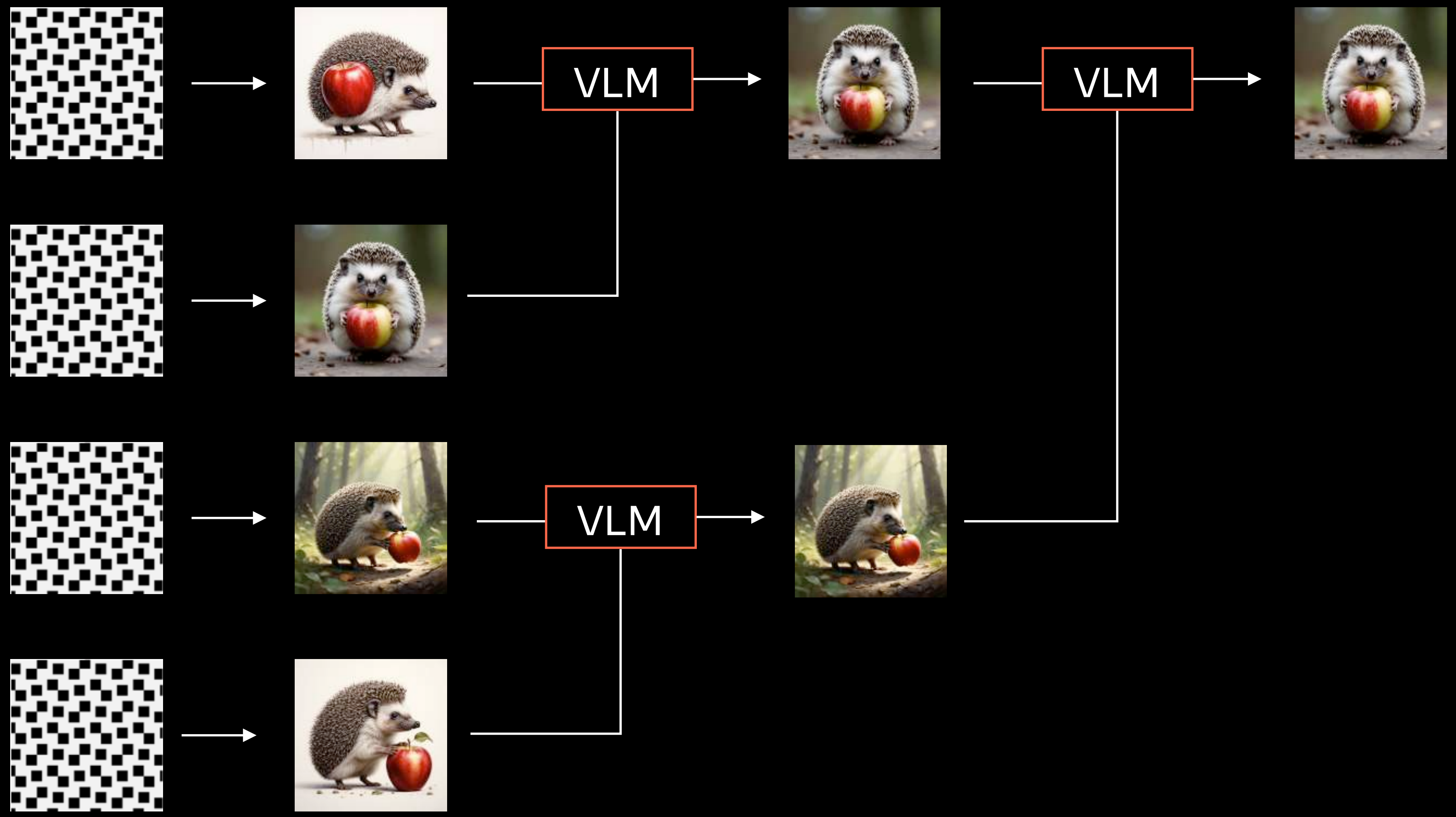
Best-on-N baseline



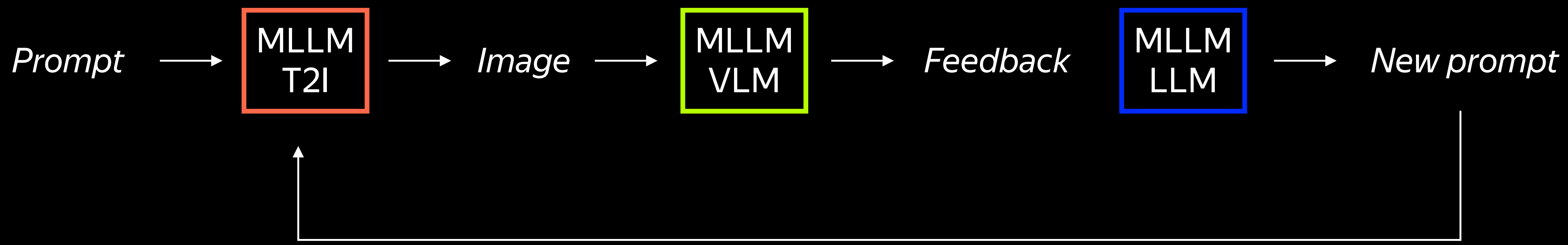
"hedgehog with an apple"

E. Xie et. al., "SANA 1.5: Efficient Scaling of Training-Time and Inference-Time Compute in Linear Diffusion Transformer", 2025

External VLM-as-judge

M. Abdul et. al., “Test-time Prompt Refinement for Text-to-Image Models“, 2025

Can multi-modal model do all that itself?



Boosts text gen and `many objects`

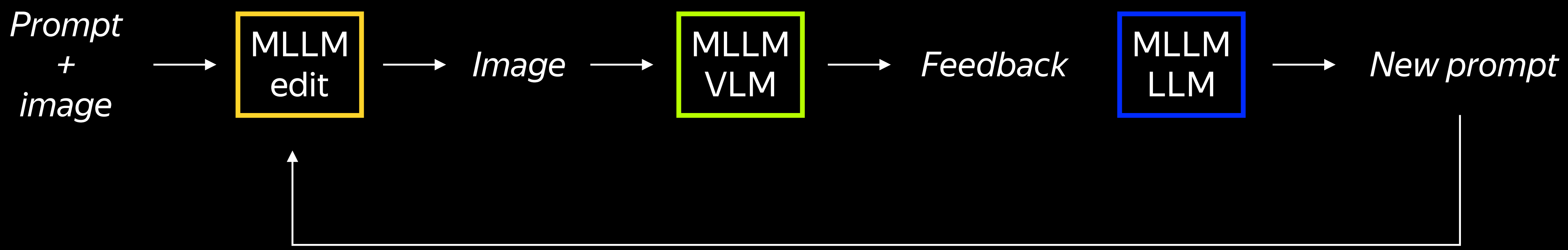
No TTS | TTS

“do an inscription `rash 435`”

“draw 5 dogs of different colors”

* preliminary results

Can we do the same for editing?



Also works pretty well

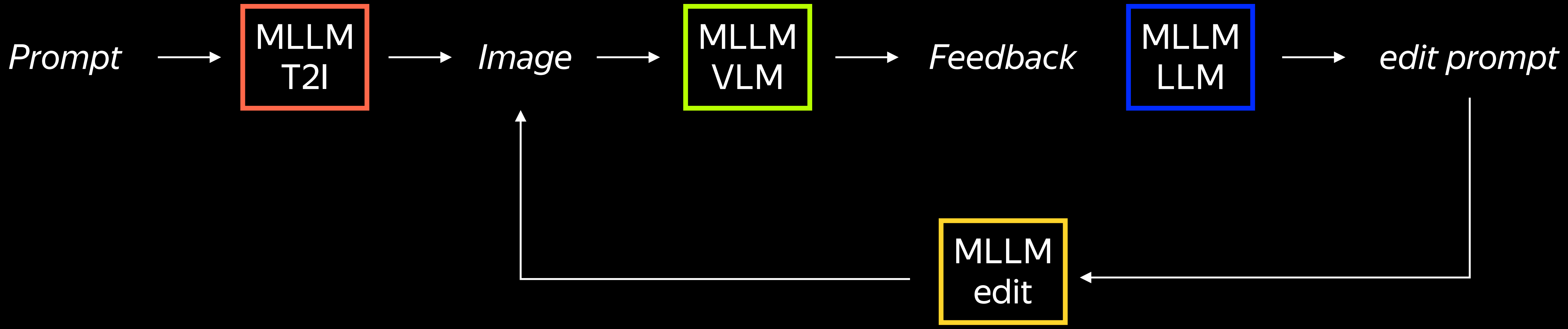
No TTS | TTS

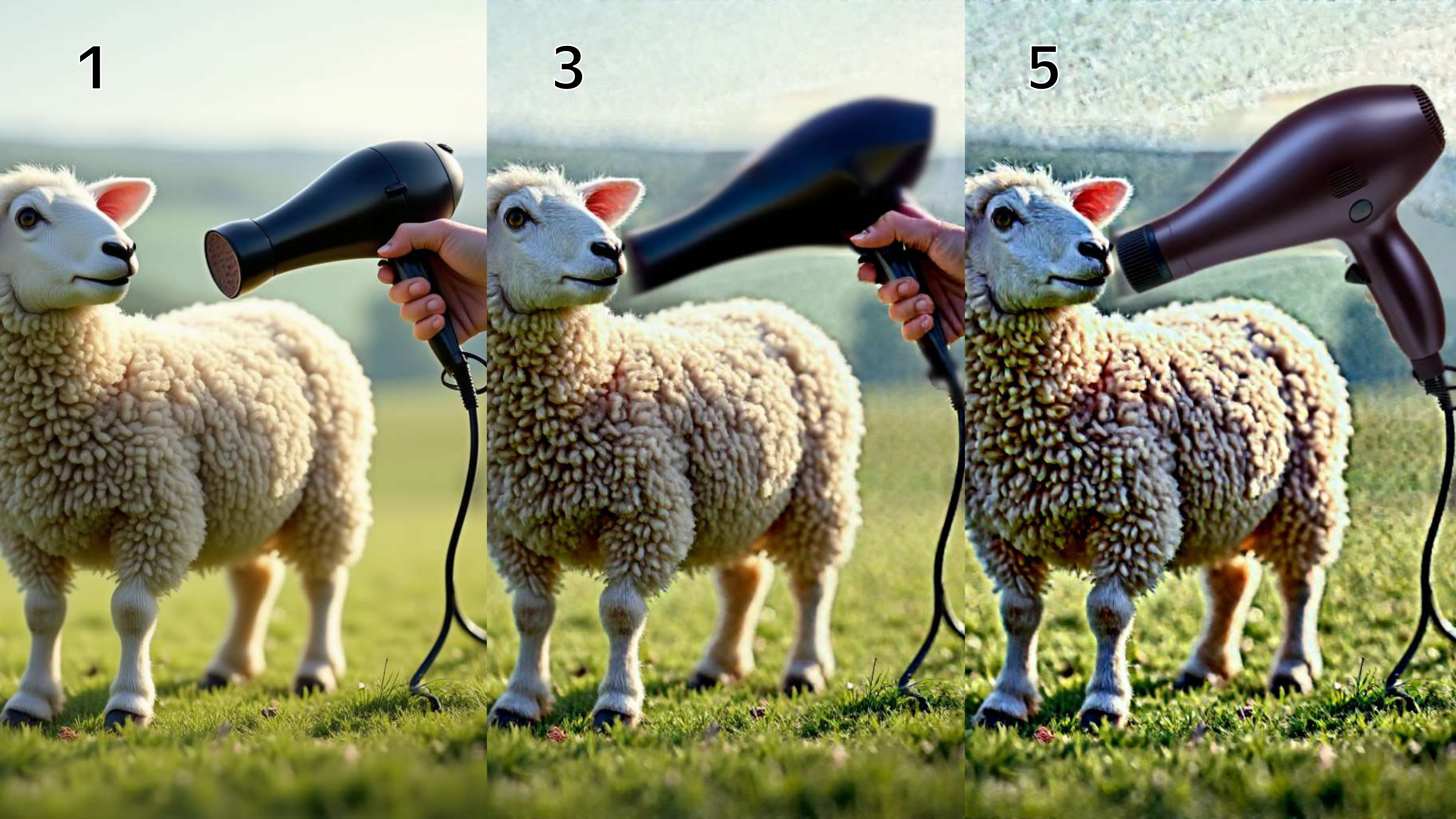
"change the action of the horses to galloping"

"Extract the person from the photo and dress them in a police uniform"

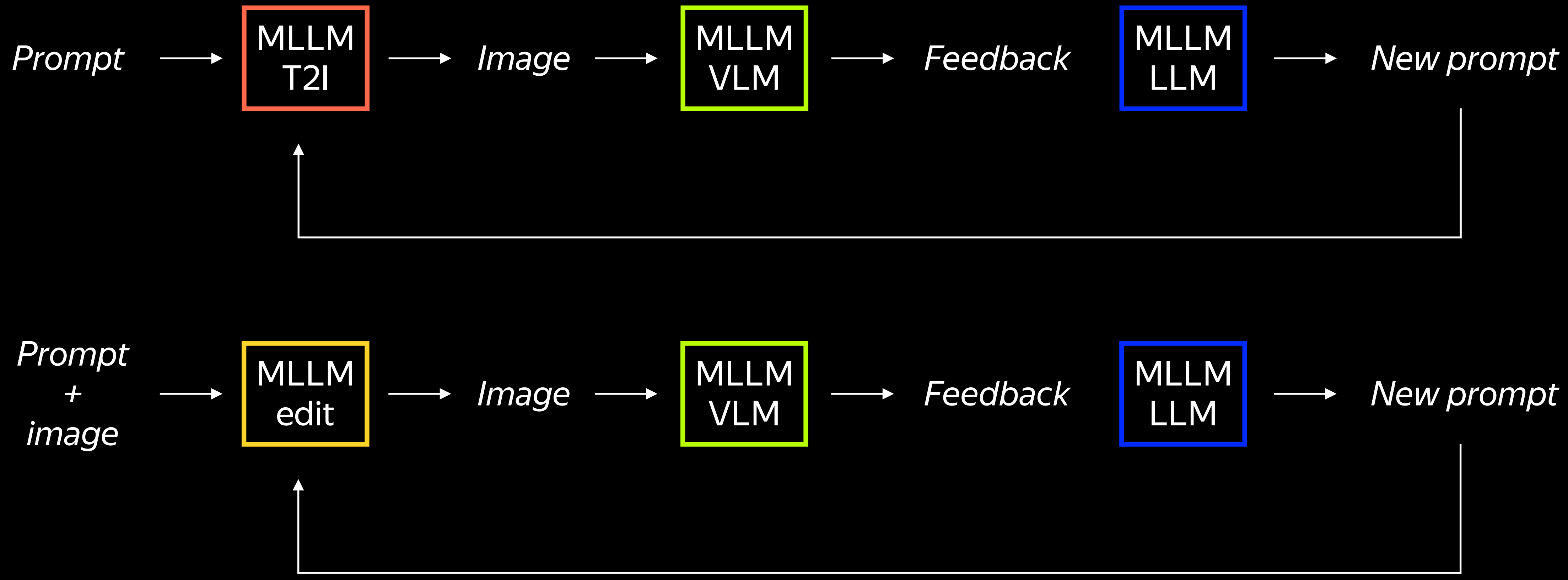
* preliminary results

Use all four tasks?




1
3
5

Distill again?



What stops us from training a reasoner on this synthetic chains?

Open Questions

Source of actual quality gain?

- VLM is knows text-image relation better than T2I?
- If so, why? (gen objective, architecture, data)
- How to enrich T2I or I2I with that?

Explicit reasoning or distill?

- Does actual reasoning emerge?
- Tasks that require visual reasoning?
- T2I to boost VLM reasoning?